

puted root to the correct one that we wanted to achieve. We will for now simply assume that it is the two functions, the given and approximating ones, that are to be close to each other. However, in Part III we will see that there are very different answers to the question of where we should apply the criteria of matching.

To summarize, we have four questions which, it will turn out, determine the formula uniquely:

- 1 What samples (function and derivative values, etc.) shall we use?
- 2 What class of functions shall we use (polynomials here in Part II)?
- 3 What criterion shall we use (exact matching, least squares, Chebyshev, etc.)?
- 4 Where shall we apply the criterion?

14.3 INTERPOLATION USING ONLY FUNCTION VALUES

Our basic tool for finding formulas is the method of undetermined coefficients. The method assumes the form of the answer, and the conditions on the formula determine the arbitrary coefficients of the form. For example, given $N + 1$ sample points (x_i, y_i) ($i = 1, 2, \dots, N + 1$), we can fit a polynomial of degree N

$$P_N(x) = a_0 + a_1x + a_2x^2 + \cdots + a_Nx^N = \sum_{k=0}^N a_k x^k$$

to the data (since the polynomial form has exactly $N + 1$ coefficients to be determined by the $N + 1$ conditions). The condition that the polynomial $P_N(x)$ pass through the sample point (x_i, y_i) is

$$y_i = P_N(x_i) = \sum_{k=0}^N a_k x_i^k \quad i = 1, 2, \dots, N + 1$$

These $N + 1$ equations determine the coefficients a_k .

The determinant of the coefficients of the unknowns, a_k , is the Vandermonde¹ determinant

$$V_{N+1} = \begin{vmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^N \\ 1 & x_2 & x_2^2 & \cdots & x_2^N \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 1 & x_{N+1} & x_{N+1}^2 & \cdots & x_{N+1}^N \end{vmatrix} = |x_i^k|$$

¹ Alexandre Theophile Vandermonde (1735–1796).

In the next section we will show that if $x_i \neq x_j$ for $i \neq j$, then the determinant cannot be zero; hence we can always solve for the a_k , and we have a solution to the interpolation problem for polynomials.

If we solve for the a_k by the use of determinants, substitute the results into the polynomial, and finally rearrange it suitably, we would have

$$\begin{vmatrix} y & 1 & x & x^2 & \cdots & x^N \\ y_1 & 1 & x_1 & x_1^2 & \cdots & x_1^N \\ y_2 & 1 & x_2 & x_2^2 & \cdots & x_2^N \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ y_{N+1} & 1 & x_{N+1} & x_{N+1}^2 & \cdots & x_{N+1}^N \end{vmatrix} = 0$$

We can see directly that this is the solution. Expanding by the elements of the top row, it is clearly a polynomial of degree N . If x and y take on the values x_i and y_i , then two rows would be the same, and hence the determinant would be zero. This observation shows that the polynomial passes through the given points.

From this form it is immediately evident that if all the y_i were zero, then the polynomial would also be identically zero, $y(x) \equiv 0$. This result is important because it means that the *interpolating polynomial is unique (within roundoff)*. The interpolating polynomial can be written in many different forms and in many different notations, but if the same information is used in two different methods, then necessarily the two resulting polynomials are the same (within roundoff). It is also easy to see what the condition is for the coefficient of x_N to be zero. We will speak of the interpolating polynomial being of degree N even if, as sometimes happens, the actual polynomial is of lower degree.

As an example of finding an interpolating polynomial consider the problem where we are given the four points $(0, 2)$, $(1, 2)$, $(2, 0)$, $(3, 0)$ and wish to determine the cubic through them (see Fig. 14.3.1). We have the four equations

$$\begin{aligned} P(0) &= 2 = a_0 \\ P(1) &= 2 = a_0 + a_1 + a_2 + a_3 \\ P(2) &= 0 = a_0 + 2a_1 + 4a_2 + 8a_3 \\ P(3) &= 0 = a_0 + 3a_1 + 9a_2 + 27a_3 \end{aligned}$$

It is easy to eliminate $a_0 = 2$. We next subtract twice the second equation from the third, and three times the second from the fourth to get

$$\begin{aligned} -2 &= 2a_2 + 6a_3 \\ -2 &= 6a_2 + 24a_3 \end{aligned}$$

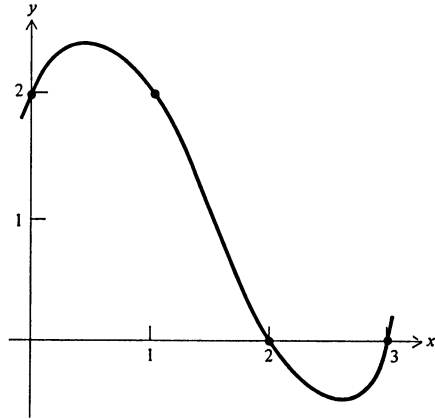


FIGURE 14.3.1

It follows that

$$4 = 6a_3 \quad \begin{cases} a_3 = 2/3 \\ a_2 = -3 \\ a_1 = 7/3 \\ a_0 = 2 \end{cases}$$

and

$$P(x) = 2 + \frac{7}{3}x - 3x^2 + \frac{2}{3}x^3$$

As a second example, we find the interpolating polynomial for

$$y = \log x \quad x = 1, 2, 3, 4$$

We have for the equations that determine the coefficients of the cubic

$$P_3 = a_0 + a_1x + a_2x^2 + a_3x^3$$

$$\log 1 = 0.0000 = a_0 + a_1 + a_2 + a_3$$

$$\log 2 = 0.3010 = a_0 + 2a_1 + 4a_2 + 8a_3$$

$$\log 3 = 0.4771 = a_0 + 3a_1 + 9a_2 + 27a_3$$

$$\log 4 = 0.6021 = a_0 + 4a_1 + 16a_2 + 64a_3$$

Eliminating a_0 pairwise to preserve some symmetry, we get

$$0.3010 = a_1 + 3a_2 + 7a_3$$

$$0.1761 = a_1 + 5a_2 + 19a_3$$

$$0.1250 = a_1 + 7a_2 + 37a_3$$

Again eliminating pairwise, this time a_1 , we get

$$-0.1249 = 2a_2 + 12a_3$$

$$-0.0511 = 2a_2 + 18a_3$$

Eliminating a_2 , we have

$$0.0738 = 6a_3 \quad \begin{cases} a_3 = 0.01230 \\ a_2 = -0.13625 \\ a_1 = 0.62365 \\ a_0 = -0.49970 \end{cases}$$

and

$$P_3(x) = 0.0123x^3 - 0.13625x^2 + 0.62365x - 0.4997$$

This, then, is the cubic approximating $y = \log x$.

PROBLEMS

14.3.1 Find a quadratic through $(0, 1)$, $(1, 2)$, and $(2, 3)$.

14.3.2 Find a quadratic through $(-1, a)$, $(0, b)$, and $(1, c)$.

$$\text{Ans. } y(x) = b + \frac{c-a}{2}x + \frac{c-2b+a}{2}x^2$$

14.3.3 Find a cubic through $(0, 1)$, $(1, 0)$, $(2, 1)$, and $(3, 0)$.

$$\text{Ans. } y(x) = \frac{-2x^3}{3} + 3x^2 - \frac{10x}{3} + 1$$

14.3.4 Find a quartic through $(-2, a)$, $(-1, b)$, $(0, c)$, $(1, b)$, and $(2, a)$. (Note that the symmetry removes the powers x and x^3 .)

14.4 THE VANDERMONDE DETERMINANT

In Sec. 14.3 we met the important Vandermonde determinant, which plays a central role, along with its variants, in the theory of polynomial approximation. The determinant is defined by

$$V_{N+1}(x_1, x_2, \dots, x_{N+1}) = |x_i^k| \quad \begin{matrix} k = 0, 1, \dots, N \\ i = 1, 2, \dots, N+1 \end{matrix}$$

Consider it as a function of the variables x_i . It is clearly a polynomial in the x_i , and a count of the exponents shows that the degree is

$$0 + 1 + 2 + 3 + \dots + N = \frac{N(N+1)}{2}$$

Now if $x_{N+1} = x_1$, then $V_{N+1} = 0$, and the determinant has the factor $x_{N+1} - x_1$. Similarly, if $x_{N+1} = x_2, x_3, \dots, x_N$, then we have the corresponding factors. Thus we have, in total, the N factors

$$\prod_{j=1}^N (x_{N+1} - x_j)$$

Again, if $x_N = x_1, x_2, \dots, x_{N-1}$, we have the $N - 1$ factors

$$\prod_{j=1}^{N-1} (x_N - x_j)$$

Repeating this argument, we see that we have all the factors

$$\prod_{k>j=1}^{N+1} (x_k - x_j)$$

This product is a polynomial of degree

$$N + (N - 1) + \dots + 1 = \frac{N(N + 1)}{2}$$

and so we have found *all* the factors, and it remains to find the multiplicative constant by which the two representations might differ. To find this constant we compare the same term in both representations. The diagonal term in the determinant is

$$1 \cdot x_2 \cdot x_3^2 \cdots x_{N+1}^N$$

The term from the left-hand sides of the product is

$$x_{N+1}^N \cdot x_N^{N-1} \cdots x_2$$

These are the same, hence

$$V_{N+1}(x_1, x_2, \dots, x_{N+1}) \equiv |x_i^k| \equiv \prod_{k>j=1}^{N+1} (x_k - x_j)$$

The Vandermonde determinant shows that the powers of x ($1, x, \dots, x^N$) are linearly independent over any set of $N + 1$ distinct points. Thus sampling a polynomial of degree N at $N + 1$ distinct points enables us to reconstruct (within roundoff) the polynomial from the samples alone. In this form it compares with the famous sampling theorem which we will examine in Chap. 34. The fundamental theorem of algebra shows that the powers of x are linearly independent in any interval; the Vandermonde shows that the first $N + 1$ powers are linearly independent over any $N + 1$ distinct points, and we know that there cannot be $N + 2$ functions linearly independent over $N + 1$ points.

PROBLEMS

14.4.1 By direct expansion of the determinant show that $V_3(x_1, x_2, x_3)$ factors properly.

14.4.2 Show directly that

$$V_{n+1} = V_n \cdot \prod_{i=1}^n (x_{n+1} - x_i)$$

and hence evaluate the Vandermonde determinant by recursion.

14.5 LAGRANGE¹ INTERPOLATION

In this section we give an alternate approach to the problem of finding the interpolating polynomial through $N + 1$ sample points. The method, ideas, and notation are more important than the result.

Suppose we first find a polynomial of degree N through the $N + 1$ points whose y values are all zero except the i th which we will take as being 1. We introduce a notation that we shall repeatedly use, namely,

$$\pi_i(x) = (x - x_1)(x - x_2) \cdots (x - x_{i-1})(x - x_{i+1}) \cdots (x - x_{N+1})$$

which is the product of all the factors *except* the i th one. Using this notation, we easily see that the solution to the problem is

$$\frac{\pi_i(x)}{\pi_i(x_i)}$$

since for $i \neq j$

$$\pi_i(x_j) = 0 \quad \pi_i(x_i) \neq 0 \quad \text{and} \quad \frac{\pi_i(x_i)}{\pi_i(x_i)} = 1$$

This *sampling polynomial* is of degree N and has the required values of being zero at all samples except the i th at which it is 1.

Now, just as we did in the first matrix inversion (Sec. 7.11), where we constructed the general solution out of the special unit solutions, so, too, we have by immediate inspection that the general solution is

$$P_N(x) = \sum_{i=1}^{N+1} y_i \left[\frac{\pi_i(x)}{\pi_i(x_i)} \right]$$

This technique of using the sampling polynomials is very useful and should be clearly understood.

¹ Joseph-Louis Lagrange (1736–1813).

The Lagrange interpolating polynomial is, of course, the same polynomial as before in Sec. 14.3, within roundoff. It is written in a different form which is more of theoretical than practical interest (since it has a great deal of roundoff in many cases).

14.6 ERROR OF POLYNOMIAL APPROXIMATIONS

Given a function $f(x)$, we took $N + 1$ sample points (x_i, y_i) , $(i = 1, 2, \dots, N + 1)$, and found a polynomial $P_N(x)$ through these points. We intend to use this polynomial in place of the original function, and it is therefore important to examine the question of how much the function and the polynomial differ at points other than the sample points (where they agree within roundoff error).

As an example, consider the function $y(x) = \log x$ (see Table 14.6). In Sec. 14.3 we found the polynomial approximating the function. As a measure of the difference $\log x - P(x)$, let us examine the values at the sample points and the midpoints between them where we would expect the error to be fairly large.

A theoretical expression for the difference between the original function $f(x)$ and the approximating polynomial $P(x)$ can be found if we observe that the difference is zero at all the sample points, and write

$$y(x) - P(x) = (x - x_1)(x - x_2) \cdots (x - x_{N+1})K(x)$$

where $K(x)$ is suitably chosen. We now choose an arbitrary x^* . We have

$$y(x^*) - P(x^*) - (x^* - x_1)(x^* - x_2) \cdots (x^* - x_{N+1})K(x^*) = 0$$

Now consider the function

$$\Phi(x) = y(x) - P(x) - (x - x_1)(x - x_2) \cdots (x - x_{N+1})K(x^*)$$

If $y(x)$ has an $(N + 1)$ st derivative, we can differentiate $N + 1$ times, and since $P(x)$ is a polynomial of degree N and $K(x^*)$ is a constant, we get

$$\Phi^{(N+1)}(x) = y^{(N+1)}(x) - (N + 1)!K(x^*)$$

Table 14.6

x	$\log x$	$P(x)$	$\log x - P(x)$
1.0	0.0000	0.0000	0.0000
1.5	0.1761	0.1707	+0.0054
2.0	0.3010	0.3010	0.0000
2.5	0.3979	0.4000	-0.0021
3.0	0.4771	0.4771	0.0000
3.5	0.5441	0.5414	+0.0027
4.0	0.6021	0.6021	0.0000

But $\Phi(x)$ vanishes $N + 2$ times (at x^* and $x_1, x_2, \dots, x_{N+1}$). Hence, by the mean value theorem, $\Phi'(x)$ vanishes at least $N + 1$ times in the interval containing all the x values (including x^*). Continuing to apply this theorem, we find that $\Phi^{(k)}(x)$ vanishes at least $N + 2 - k$ times and that $\Phi^{(N+1)}(x)$ vanishes at least once. Thus there is an $\bar{x}$ in the interval of the x values such that

$$y^{(N+1)}(\bar{x}) = (N + 1)!K(x^*)$$

We now have a value for the constant $K(x^*)$ and can put this back in the original expression to get

$$y(x^*) = P(x^*) + \frac{(x^* - x_1)(x^* - x_2) \cdots (x^* - x_{N+1})y^{(N+1)}(\bar{x})}{(N + 1)!}$$

But, since x^* was arbitrary, we may write x^* as x , and we have finally

$$y(x) = P(x) + \frac{(x - x_1)(x - x_2) \cdots (x - x_{N+1})y^{(N+1)}(\bar{x})}{(N + 1)!}$$

If we use this error term to estimate the error made in interpolating in the log table, we get

$$\frac{(x - 1)(x - 2)(x - 3)(x - 4)}{4!} y^{(4)}(\bar{x})$$

To estimate the error at $x = 3/2$, we obtain

$$\frac{(1/2)(-1/2)(-3/2)(-5/2)}{4!} \left(\frac{-6}{\bar{x}^4} \right) = \frac{15}{64} \frac{1}{\bar{x}_4}$$

All that we know is that $1 \leq x \leq 4$. At worst the error is $15/64 \approx 0.23$, while at best it is 0.001—from Table 14.6 it is about 0.0054, which shows how little accuracy such an error term actually gives in this case.

It should be noted that the value $\bar{x} = \bar{x}(x)$ depends on x , and there may be several $\bar{x}$'s for some values of x . Also, $\bar{x}$ is not necessarily a continuous function of x . This latter effect is illustrated by applying the mean value theorem to the function $y(x) = x(1 - x)^2$ (Fig. 14.6.1) and choosing $a = 0$ in the usual form for the mean value theorem:

$$y'(\bar{x}) = \frac{y(x) - y(a)}{x - a} = \frac{y(x)}{x} = (1 - x)^2$$

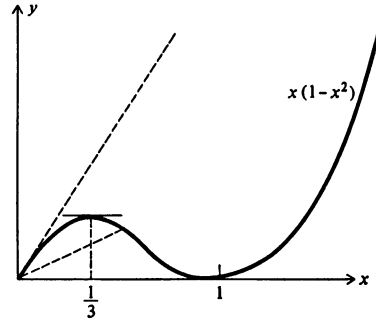


FIGURE 14.6.1

Thus $y'(\bar{x}) \geq 0$, and as

x goes to 0 to 1, $\bar{x}$ goes from 0 to $1/3$

x goes from 1 to 2, $\bar{x}$ goes from $1/3$ to 0

x goes beyond 2, $\bar{x}$ jumps over the interval where $y' < 0$, to $x > 1$

Hence $\bar{x} = \bar{x}(x)$ is not a continuous function of x .

14.7 DIFFICULTY OF POLYNOMIAL APPROXIMATION

It is customary, as in the previous section, to express the error of a polynomial approximation in terms of a suitable derivative of the function being approximated. It is usually thought that for "most" reasonable functions such expressions for the error become small for sufficiently large n , but this is not so.

Suppose that we restrict our discussion to analytic functions, that is, functions having a convergent Taylor series

$$y(x) = \sum_{n=0}^{\infty} \frac{(x - x_0)^n}{n!} y^{(n)}(x_0)$$

at all points x_0 in our range of interest. If, further, the function is an integral (entire) function, that is, converges everywhere in the finite part of the complex plane, as do $\sin x$, e^x , polynomials in x , etc., then it is indeed possible that all the higher derivatives are small. But if the function has a singularity in the finite part of the complex plane, as do $\tan x$, $\log x$, rational functions in x , etc., then the Taylor series must have a finite radius of convergence R , and this in turn means that for an infinite number of values of n

$$\frac{(R + \varepsilon)^n}{n!} |y^{(n)}(x_0)| \geq 1 \quad \varepsilon > 0$$

$$|y^{(n)}(x_0)| \geq \frac{n!}{(R + \varepsilon)^n}$$

In words, the upper bound on the n th derivative grows as $n!$. As an example, consider

$$\begin{aligned}
 y &= \ln x \\
 y' &= \frac{1}{x} \\
 y'' &= \frac{-1}{x^2} \\
 y''' &= \frac{2!}{x^3} \\
 &\dots\dots\dots \\
 y^{(n)} &= \frac{(-1)^{n-1}(n-1)!}{x^n}
 \end{aligned}$$

Thus, even though the curve $y = \ln x$ looks smooth near some value x , nevertheless as n gets large, the derivatives at this point become very large in size and tend to behave as $n!$ or worse.

This is the general case; for “most” functions some of the higher-order derivatives tend to grow as $n!$. It is only for certain integral functions that all the derivatives can remain bounded.¹ Even for polynomials there is a tendency for the derivatives to grow in size until the N th one which is $a_0 N!$, after which they suddenly all become zero. Of course, many of the higher derivatives of a function may be small while some of the others are large. For example, an even function, $f(x^2)$, has all its odd-order derivatives equal to zero at the origin, but if the function is not an integral function, then there are also an infinite number of the even-order derivatives which tend to behave as $n!$, or worse.

It would be helpful if we had to deal only with integral functions which can have nicely bounded derivatives, but the facts seem to be that if the function is an integral function, quite likely the whole problem can be solved analytically, whereas if it is necessary to use numerical methods, the function is likely to be rather poorly behaved.

The Weierstrass theorem which states, loosely speaking, that a continuous function in a closed interval can be uniformly approximated by a polynomial is often cited as justification for using a polynomial approximation [49, pp. 28–31].

¹ The converse is not true; the derivatives of an integral function do not need to be bounded. For example, if $y = xe^x$, then $y^{(n)}(0) = n$.

However, the method that we have been using of exact matching at the sample points is not the way in which the Weierstrass polynomial is defined; hence the theorem, while possibly suggestive, does not apply. Indeed, the simple function [56]

$$y(x) = \frac{1}{1+x^2} \quad -5 \leq x \leq 5$$

is well known as a bad example, since as a selected set of equally spaced points is increased in number, the approximating polynomial (in the sense of exact matching) diverges from the function between some of the sample points. Thus, even for equally spaced sample points, we cannot rely on a polynomial to be a good approximation if exact matching at the sample points is the criterion used to select the polynomial. The explanation of this phenomenon is, of course, that the derivatives grow too rapidly.

As an example of this effect, consider the Riemann zeta function given in Table 10.6.2 in Sec. 10.6. The top line of differences does not tend to become small very rapidly, which is caused by the obvious singularity at $x = 1$.

It should be noted, however, that if a table is given at one-half the spacing of a second table of the same function, then the first differences are one-half as large, the second one-fourth as large, etc., as those of the second table. This suggests an empirical rule: If the differences of a table approach zero rapidly, then probably too small a spacing was used and the table was overcomputed, whereas if the differences do not become small, then a finer spacing should be considered.

A final, and perhaps much more basic, objection to polynomial approximation is that it rarely has any physical implications that will lead to useful insights. Remember: The purpose of computing is insight, not numbers.

On the other hand, the theory is simple and well developed, requires a minimum of computation, and is useful. Experience shows that polynomials often do a good job, although the error term may be either unobtainable or pessimistic. The values of the difference table can be highly suggestive of the values of the derivative, and they indicate the smallness of the contribution due to taking more terms in the Newton interpolation polynomial. However, the use of the differences as a guide to the values of the derivatives can be dangerous; thus, for integer x the values of $\sin \pi x$ are all zero, so that the difference table is also all zeros, suggesting zero error in the approximation

$$\sin \pi x = 0$$

which is hardly true. The fallacy is obvious here, but in some situations it might be overlooked.

PROBLEMS

14.7.1 Compute the n th derivative of

$$\frac{1}{1+x^2} \equiv \frac{1}{2i} \left(\frac{1}{x-i} - \frac{1}{x+i} \right) \quad i = \sqrt{-1}$$

and evaluate at $x = 0$.

14.8 ON SELECTING SAMPLE POINTS

The problem of which sample points to select occurs when we try to interpolate. If we had complete freedom in choosing, we would select the value x for which we were going to interpolate, thus making the problem trivial. But it often happens that we have an extensive set of values (x_i, y_i) which we might use, none of which coincides with the desired value x . In the absence of knowledge of the size of the derivative occurring in the error term we can only pick our samples to minimize the factor

$$(x - x_1)(x - x_2) \cdots (x - x_{n+1})$$

in front of the derivative term.

Common sense and the minimization of this factor coincide, that is, use information close to the place at which we want to interpolate.

14.9 SUBTABULATION

Subtabulation means interpolating values into a table at equally spaced points. This is often far cheaper than computing the values at the points. Of course it requires the use of some standard subtabulating routine, which is additional programming, but it can often save machine time. For example, Table 14.9 for cubic interpolation gives the values to use to multiply the original table values to get the subtabulated values at one-tenth the spacing. Other formulas are easily found.

Since the results of a computation are often to be used by humans, it is necessary to have the final results at a fine enough spacing to support linear interpolation (most humans will not go into quadratic or cubic interpolation themselves). When the computation of the individual values is very expensive, that is the time to consider subtabulation, a highly developed field due to past hand-calculation efforts of table makers. Before making any table, consult Fox [12].

Table 14.9

$$f(x) = A_{-1}(x)f(-1) + A_0(x)f(0) + A_1(x)f(1) + A_2(x)f(2)$$

x	A_{-1}	A_0	A_1	A_2	
0.0	0	1.0	0.0	0.0	1.0
0.1	-0.0285	0.9405	0.1045	-0.0165	0.9
0.2	-0.0480	0.8640	0.2160	-0.0320	0.8
0.3	-0.0595	0.7735	0.3315	-0.0455	0.7
0.4	-0.0640	0.6720	0.4480	-0.0560	0.6
0.5	-0.0625	0.5625	0.5625	-0.0625	0.5
	A_2	A_1	A_0	A_{-1}	x

PROBLEMS

- 14.9.1 Using these Lagrange interpolating coefficients, compute the error integral at 1.40 from Table 10.6.3.
- 14.9.2 Using Table 10.6.3 of the error integral (Sec. 10.6), compute the entries in the table for the midpoints.

15.1 INTRODUCTION

The purpose of this chapter is to present a uniform method for finding a wide class of formulas based on sample values of the function. Chapters 17, 18, and 19 will elaborate the basic method for formulas that use other kinds of information besides the function values. Chapter 16 is devoted to a uniform method for finding the corresponding error terms of the formulas.

The emphasis is on a *uniform* approach to the problem of finding a formula to meet a given situation. We are not interested in giving an exhaustive list of formulas because, among other reasons, it is hopeless. The situation is something like the problem of trigonometric identities. There are so many possible identities that after giving a short list of basic ones and assuming a modest amount of experience in deriving others, we drop the subject. One does not seriously propose to make a large table of trigonometric identities—it is generally easier to derive the one you need from a few basic ones than to look it up in some large table. Similarly, after giving a few standard formulas, we shall be content with a description of *how* to find any reasonable formula you might want from the given class of possible formulas.

The problem may be viewed as one of information retrieval; we wish to find a particular piece of information, not knowing for sure that it has ever been recorded, and instead of trying to provide a catalog of known results (which would necessarily be finite and to a great extent represent past interests), we prefer a method of creating it as needed. *Instead of information retrieval we prefer information regeneration.* Of course this attitude will not work in all areas of knowledge, but in a mature science the principles should show how to recreate the information on demand. We will end these few chapters with a flow diagram indicating how to derive formulas. Currently on computers we recreate the elementary, special function values when needed rather than use a table, and so the approach is not novel.

Our basic approach is through these four questions (given in the preceding chapter):

- 1 What samples?
- 2 What class of functions?
- 3 What criterion of goodness of fit?
- 4 Where is the test to be applied?

Here in Part II we use polynomials as our approximating functions. This, as we said before, is the classical approach. In Parts III and IV we will use other classes.

In the next few chapters we will use the criterion of the formula being true for the successive powers of x , as high as we can go. Later we will try other criteria. We will also apply the test of accuracy to the functions themselves, and only much later will we consider other ways of applying the criteria of accuracy.

Returning to the first question, What samples?, we will confine ourselves in this chapter to function values, in Chap. 17 to function values plus various orders of derivatives of the functions, in Chap. 18 to function values and their differences, and in Chap. 19 we shall take the location of the samples as parameters which gives further freedom in the choice of a formula to fit a given situation.

Thus, the four questions provide the structure of the chapters and the basis for our uniform method which is used throughout much of the book.

15.2 FORMULAS USING INTERPOLATION

The classical approach to finding a formula for some operation, typically integration, is to first find the interpolating polynomial and then apply the operation, typically integration, to the polynomial. This is the method of *analytic substitution*.

As an example, consider finding the formula for integrating a function between the two given endpoints a and b . We have the two points $(a, y(a))$ and $(b, y(b))$, and the interpolating line through them is

$$y(x) = \frac{(b-x)y(a) + (x-a)y(b)}{b-a}$$

Integrating this curve, we get

$$\begin{aligned} \int_a^b y(x) dx &= \int_a^b \frac{(b-x)y(a) + (x-a)y(b)}{b-a} dx \\ &= \frac{-(b-x)^2 y(a) + (x-a)^2 y(b)}{2(b-a)} \Big|_a^b \\ &= \frac{(b-a)^2 y(b) + (b-a)^2 y(a)}{2(b-a)} \\ &= \left[\frac{y(b) + y(a)}{2} \right] (b-a) \end{aligned}$$

which is the well-known trapezoid rule.

As a second example, consider finding a quadratic through three equally spaced points and integrating to find the area. We take the points $(-1, y(-1))$, $(0, y(0))$, and $(1, y(1))$. Using undetermined coefficients, we assume the form

$$\begin{aligned} y &= A_0 + A_1 x + A_2 x^2 \\ y(-1) &= A_0 - A_1 + A_2 \\ y(0) &= A_0 \\ y(1) &= A_0 + A_1 + A_2 \end{aligned}$$

It follows easily that

$$\begin{aligned} A_0 &= y(0) \\ A_1 &= \frac{1}{2}[y(1) - y(-1)] \\ A_2 &= \frac{1}{2}[y(1) - 2y(0) + y(-1)] \end{aligned}$$

The integral of this quadratic is

$$\begin{aligned} I &= \int_{-1}^1 (A_0 + A_1 x + A_2 x^2) dx = 2A_0 + \frac{2A_2}{3} \\ &= 2y_0 + \frac{1}{3}[y(1) - 2y(0) + y(-1)] \\ &= \frac{1}{3}[y(-1) + 4y(0) + y(1)] \end{aligned}$$

which is the well-known Simpson's formula.

Suppose we apply this formula to

$$\int_0^1 e^x dx$$

We need to shift either the formula or the integral so that the ranges match; we move the formula

$$\begin{array}{rcl} \int_0^1 y dx & = & \frac{1}{6}[y(0) + 4y(\frac{1}{2}) + y(1)] \\ e^0 & = & 1.0000 \qquad 1 \cdot e^0 = 1.0000 \\ e^{1/2} & = & 1.6487 \qquad 4 \cdot e^{1/2} = 6.5948 \\ e^1 & = & 2.7183 \qquad 1 \cdot e^1 = 2.7183 \\ & & \hline & & 10.3131 \end{array}$$

$$I_{\text{calc}} = \frac{1}{6}(10.3131) = 1.7188$$

$$I_{\text{anal}} = \int_0^1 e^x dx = e^1 - 1 = 1.7183 \quad \text{error} = 0.0005$$

PROBLEMS

15.2.1 Compute $\int_0^1 e^{-x} dx$ using Simpson's formula.

15.2.2 Compute $\int_0^\pi \sin x dx$ using Simpson's formula.

15.2.3 Find the midpoint formula $\int_0^1 f(x) dx = f(1/2)$

15.3 THE TAYLOR-SERIES METHOD OF FINDING FORMULAS

There is a second method of finding formulas that is occasionally used. In this method the function, typically an integrand, is expanded into a Taylor series about the midpoint (usually), and then the unknown coefficients are determined so that as many terms of the series exactly cancel as possible; the rest is regarded as the error, the *truncation error*. Thus the truncation error is the rest of the infinite series that did not exactly cancel out.

As an example of the Taylor-series approach, consider again finding Simpson's formula.

$$\int_{-1}^1 f(x) dx = af(-1) + bf(0) + cf(1)$$

We write $f(x)$ as a Taylor series about $x = 0$

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{(4)}(0) + \cdots$$

We get on the two sides of the formula

$$\begin{aligned} & 2f(0) + \frac{2}{3!}f''(0) + \frac{2}{5!}f^{(5)}(0) + \cdots \\ &= a \left[f(0) - f'(0) + \frac{f''(0)}{2!} - \frac{f'''(0)}{3!} + \frac{f^{(4)}(0)}{4!} - \cdots \right] \\ & \quad + b[f(0)] \\ & \quad + c \left[f(0) + f'(0) + \frac{f''(0)}{2!} + \frac{f'''(0)}{3!} + \frac{f^{(4)}(0)}{4!} + \cdots \right] \end{aligned}$$

Since we want this formula to be true for a wide class of functions, we must regard the derivative values at $x = 0$ as being independent variables, and we therefore set their coefficients equal to zero. This gives the equations [using $k!f^{(k)}(0)$]

$$\begin{aligned} 2 &= a + b + c \\ 0 &= -a + c \\ \frac{2}{3} &= a + c \\ 0 &= -a + c \\ \frac{2}{5} &= a + c + \text{error} \end{aligned}$$

The solution of these equations is $a = 1/3 = c$ and $b = 4/3$, as before. We see that Simpson's formula is exactly true for cubics even though it was originally

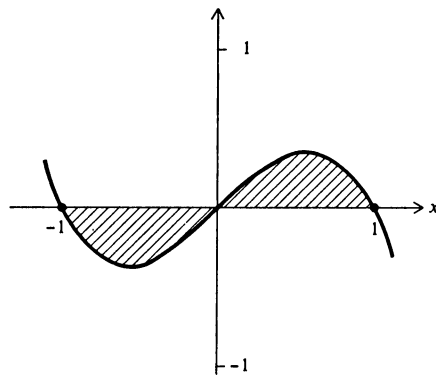


FIGURE 15.3.1

derived only for quadratics. This is true because due to symmetry the cubic part of the function *exactly* cancels out. Thus the cubic

$$y(x) = x(1 - x^2)$$

which vanishes at the sample points has its area equal to zero, and any multiple of this can be subtracted from a function without changing the answer (see Fig. 15.3.1).

The value of the error E_4 is, from the last equation,

$$E_4 = \frac{2}{3} - \frac{2}{3} = -\frac{4}{15}$$

15.4 THE DIRECT METHOD OF FINDING FORMULAS

Instead of finding formulas from the interpolating polynomial, which is the classic method, we shall adopt another method, the direct approach to the formula. This method is not only, as its name implies, more direct, it is also more powerful in that it can find formulas that the interpolation method cannot—by going directly to the formula, we can avoid those cases in which there is no interpolating polynomial but for which there is a formula of the required form. But far more important is the fact that the direct method provides a natural basis for generalization to other classes of functions. It will be our basic approach to finding formulas. For polynomials, the direct method resembles the Taylor-series method, but has a different approach to the error.

To illustrate the direct method, suppose we again find Simpson's formula. We want a formula for the integral that uses the function values $f(-1)$, $f(0)$, and $f(1)$

$$\int_{-1}^1 f(x) dx = af(-1) + bf(0) + cf(1)$$

where a , b , and c , are the undetermined coefficients. We make the formula exactly correct for $f(x)$ equals to 1, x , and x^2 .

$$f(x) = 1: \quad 2 = a + b + c \quad (15.4.1)$$

$$f(x) = x: \quad 0 = -a + c \quad (15.4.2)$$

$$f(x) = x^2: \quad \frac{2}{3} = a + c \quad (15.4.3)$$

These equations are exactly those of the Taylor-series approach, and so the answer is the same.

We will, for the moment, rate formulas by the error found from the first higher power of x that does not exactly cancel; in Chap. 16 we will in part justify this approach. Putting x^3 in the formula, we get

$$0 = -a + c$$

which is true from Eq. (15.4.2).

Using x^4 , we get

$$\begin{aligned}\frac{2}{5} &= a + c + E_4 \\ E_4 &= \frac{2}{5} - \frac{1}{3} - \frac{1}{3} = -\frac{4}{15}\end{aligned}$$

as we should.

As a second example, consider finding a formula of the form

$$\int_{-1}^1 f(x) \sin \frac{\pi}{2} x \, dx = w_{-1} f(-1) + w_0 f(0) + w_1 f(1)$$

With three parameters w_{-1} , w_0 , and w_1 we can make the formula exact for $f(x)$ equal to 1, x , and x^2 . We get

$$\begin{aligned}0 &= w_{-1} + w_0 + w_1 \\ \frac{8}{\pi^2} &= -w_{-1} + w_1 \\ 0 &= w_{-1} + w_1\end{aligned}$$

where we have used

$$\int_{-1}^1 \sin \frac{\pi}{2} x \, dx = 0 \quad \int_{-1}^1 x \sin \frac{\pi}{2} x \, dx = \frac{8}{\pi^2} \quad \int_{-1}^1 x^2 \sin \frac{\pi}{2} x \, dx = 0$$

The solution of the three equations is

$$w_1 = \frac{4}{\pi^2} = -w_{-1} \quad w_0 = 0$$

and the formula is

$$\int_{-1}^1 f(x) \sin \frac{\pi}{2} x \, dx = \frac{4}{\pi^2} [f(1) - f(-1)]$$

If we try $f(x) = x^3$, we get (on setting $x = \frac{2}{\pi} y$)

$$\begin{aligned}\int_{-1}^1 x^3 \sin \frac{\pi}{2} x \, dx &= \left(\frac{2}{\pi}\right)^4 \int_{-\pi/2}^{\pi/2} y^3 \sin y \, dy \\ &= \left(\frac{2}{\pi}\right)^4 [(3y^2 - 6)\sin y - (y^3 - 6y)\cos y] \Big|_{-\pi/2}^{\pi/2} \\ &= \left(\frac{2}{\pi}\right)^4 \left[3\left(\frac{\pi}{2}\right)^2 - 6\right] 2 = \frac{24}{\pi^4}(\pi^2 - 8) = E_3 + \frac{8}{\pi^2}\end{aligned}$$

Hence E_3 measures the error.

As a final example of the direct method, consider a formula of the form

$$\int_{-1}^1 f(x) dx = w_{-2/3} f(-2/3) + w_0 f(0) + w_{2/3} f(2/3)$$

We try successive powers of x

$$f(x) = 1: \quad 2 = w_{-2/3} + w_0 + w_{2/3} \quad (15.4.4)$$

$$f(x) = x: \quad 0 = -\frac{2}{3}w_{-2/3} + \left(\frac{2}{3}\right)w_{2/3} \quad (15.4.5)$$

$$f(x) = x^2: \quad \frac{2}{3} = \left(-\frac{2}{3}\right)^2 w_{-2/3} + \left(\frac{2}{3}\right)^2 w_{2/3} \quad (15.4.6)$$

$$f(x) = x^3: \quad 0 = \left(-\frac{2}{3}\right)^3 w_{-2/3} + \left(\frac{2}{3}\right)^3 w_{2/3} \quad (15.4.7)$$

$$f(x) = x^4: \quad \frac{2}{5} = \left(\frac{2}{3}\right)^4 w_{-2/3} + \left(\frac{2}{3}\right)^4 w_{2/3} + E_4 \quad (15.4.8)$$

From Eq. (15.4.5)

$$w_{2/3} = w_{-2/3}$$

Then from Eqs. (15.4.4) and (15.4.6)

$$2 = 2w_{2/3} + w_0$$

$$1 = \frac{2}{3} \cdot 2w_{2/3}$$

we get

$$w_{2/3} = 3/4 = w_{-2/3}$$

$$w_0 = 2/4$$

Equation (15.4.7) follows from Eq. (15.4.5), and Eq. (15.4.8) gives

$$E_4 = \frac{14}{135}$$

These three examples should make clear the direct method; there may, as in the second case, be some difficult integrations to get some numbers, but the method is simple.

PROBLEMS

15.4.1 Find the formula for using

$$\int_{-\pi}^{\pi} f(x) \sin x dx$$

using $f(-\pi)$, $f(-\pi/2)$, $f(0)$, $f(\pi/2)$, and $f(\pi)$.

$$\text{Ans. } \int_{-\pi}^{\pi} f(x) \sin x dx = \left(1 - \frac{8}{\pi^2}\right) [f(\pi) - f(-\pi)] + \frac{16}{\pi^2} [f(\pi/2) - f(-\pi/2)]$$

15.4.2 Find a formula for

$$-\int_0^1 f(x) \ln x \, dx$$

using $f(0)$, $f(1/2)$, and $f(1)$.

$$\text{Ans. } -\int_0^1 f(x) \ln x \, dx = \frac{1}{36} [17f(0) + 20f(1/2) - f(1)] \quad E_3 = 1/48$$

15.4.3 Find $\int_0^1 f(x) \, dx = w_0 f(1/3) + w_1 f(2/3)$

15.4.4 Find $\int_0^\infty e^{-x} f(x) \, dx = w_0 f(0) + w_2 f(2)$

15.4.5 Find $-\int_0^1 f(x) \ln x \, dx = w_0 f(0) + w_1 f(1/3) + w_2 f(2/3) + w_3 f(1)$

15.5 THE INVERSE VANDERMONDE

The method illustrated in the last section consists of the following steps:

- 1 Write down the proposed formula with undetermined coefficients.
- 2 For $f(x)$ substitute $1, x, x^2, x^3, \dots$ until there are as many equations as there are coefficients—later we may find that we need more equations.
- 3 Solve the resulting linear equations.

The equations can always be solved since the determinant of the unknowns is the Vandermonde determinant, which we proved (Sec. 14.4) cannot be zero if the sample points are distinct.

The equations define the formula and will be called the *defining equations*.

If we write the matrix of unknowns as

$$X = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ x_1 & x_2 & x_3 & \cdots & x_N \\ x_1^2 & x_2^2 & x_3^2 & \cdots & x_N^2 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_1^{N-1} & x_2^{N-1} & x_3^{N-1} & \cdots & x_N^{N-1} \end{pmatrix}$$

and write the *weights*, which we have generally been labeling w_i , as a column vector w , then the left-hand sides are the *moments* of the operator for which we are finding a formula and are the column vector m in the equation

$$m = Xw$$

From this we see that there is a separation of the sample points used and the operator. The sample points occur only in the matrix, while the moments of the

operator, which are all that we ever use of the operator in the method, are the resultant vector. If we could find in some direct fashion the inverse matrix X^{-1} , which depends *only* on the sample points, then questions of the loss of accuracy in the solution process would be easy to examine. We therefore set about to invert the matrix corresponding to the Vandermonde determinant.

The basis for the inversion process is the *sample polynomials* introduced in the Lagrange interpolation process (Sec. 14.5), namely,

$$\frac{\pi_i(x)}{\pi_i(x_i)}$$

Notice that

$$\frac{\pi_i(x_j)}{\pi_i(x_i)} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

We will work with the $\pi_i(x)$ which have the property

$$\pi_i(x_j) = \begin{cases} 0 & i \neq j \\ \pi_i(x_i) \neq 0 & i = j \end{cases}$$

Expanding out the $\pi_i(x)$, we have the polynomials

$$\pi_i(x) = c_{i,N-1}x^{N-1} + c_{i,N-2}x^{N-2} + \cdots + c_{i,0} \quad c_{i,N-1} = 1$$

where the $c_{i,j}$ are the elementary symmetric functions of the sample points x_j *except* for the i th sample point. The matrix of coefficients times the Vandermonde matrix gives

$$\begin{pmatrix} c_{1,0} & c_{1,1} & c_{1,2} & \cdots & c_{1,N-1} \\ c_{2,0} & c_{2,1} & c_{2,2} & \cdots & c_{2,N-1} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ c_{N,0} & c_{N,1} & c_{N,2} & \cdots & c_{N,N-1} \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ x_1 & x_2 & x_3 & \cdots & x_N \\ x_1^2 & x_2^2 & x_3^2 & \cdots & x_N^2 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_1^{N-1} & x_2^{N-1} & x_3^{N-1} & \cdots & x_N^{N-1} \end{pmatrix} = (\pi_i(x_j))$$

which for our initial purposes is close enough to the inverse. To get the inverse, evidently we only need to divide the $c_{i,j}$ by $\pi_i(x_i)$, or else we can divide the answer in the i th position by $\pi_i(x_i)$ to get w_i . Let C be the inverse matrix

$$C = X^{-1}$$

then

$$w = Cm$$

is the solution of the system of defining equations.

It is important to notice that the inverse is found by forming the elementary symmetric functions of the sample points, excluding one sample point each time, and using these as the coefficients of the inverse, *provided* they are normalized by the constant $\pi_i(x_i)$. There are no awkward questions of the solution of the system of equations and ill-conditioning of the matrix; the method is fairly straightforward and should be clearly understood since it will be developed further in later chapters.

15.6 UNIVERSAL MATRICES

The special case of equally spaced samples arises frequently in practice, and it is worth tabulating these *universal matrices*. We have labeled them S_n with the sample points used in the following parentheses; thus $S_4(-3/2, -1/2, 1/2, 3/2)$ means that four sample points at $-3/2, -1/2, 1/2$, and $3/2$ are used. When this universal matrix is multiplied by the vector of moments of the operator we are approximating, we get the weights of the formula.

Universal Matrices

$$\begin{aligned}
 S_2 &= S_2(-\tfrac{1}{2}, \tfrac{1}{2}) = \begin{pmatrix} \tfrac{1}{2} & -1 \\ \tfrac{1}{2} & 1 \end{pmatrix} = \tfrac{1}{2} \begin{pmatrix} 1 & -2 \\ 1 & 2 \end{pmatrix} \\
 S_3 &= S_3(-1, 0, 1) = \begin{pmatrix} 0 & -\tfrac{1}{2} & \tfrac{1}{2} \\ 1 & 0 & -1 \\ 0 & \tfrac{1}{2} & \tfrac{1}{2} \end{pmatrix} = \tfrac{1}{2} \begin{pmatrix} 0 & -1 & 1 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix} \\
 S_4 &= S_4(-\tfrac{3}{2}, -\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{3}{2}) = \tfrac{1}{48} \begin{pmatrix} -3 & 2 & 12 & -8 \\ 27 & -54 & -12 & 24 \\ 27 & 54 & -12 & -24 \\ -3 & -2 & 12 & 8 \end{pmatrix} \\
 S_5 &= S_5(-2, -1, 0, 1, 2) = \tfrac{1}{840} \begin{pmatrix} 0 & 2 & -1 & -2 & 1 \\ 0 & -16 & 16 & 4 & -4 \\ 24 & 0 & -30 & 0 & 6 \\ 0 & 16 & 16 & -4 & -4 \\ 0 & -2 & -1 & 2 & 1 \end{pmatrix} \\
 S_6 &= S_6(-\tfrac{5}{2}, -\tfrac{3}{2}, \dots, \tfrac{5}{2}) \\
 &= \tfrac{1}{3,840} \begin{pmatrix} 45 & -18 & -200 & 80 & 80 & -32 \\ -375 & 250 & 1,560 & -1,040 & -240 & 160 \\ 2,250 & -4,500 & -1,360 & 2,720 & 160 & -320 \\ 2,250 & 4,500 & -1,360 & -2,720 & 160 & 320 \\ -375 & -250 & 1,560 & 1,040 & -240 & -160 \\ 45 & 18 & -200 & -80 & 80 & 32 \end{pmatrix} \\
 S_7 &= \tfrac{1}{7,120} \begin{pmatrix} 0 & -12 & 4 & 15 & -5 & -3 & 1 \\ 0 & 108 & -54 & -120 & 60 & 12 & -6 \\ 0 & -540 & 540 & 195 & -195 & -15 & 15 \\ 720 & 0 & -980 & 0 & 280 & 0 & -20 \\ 0 & 540 & 540 & -195 & -195 & 15 & 15 \\ 0 & -108 & -54 & 120 & 60 & -12 & -6 \\ 0 & 12 & 4 & -15 & -5 & 3 & 1 \end{pmatrix}
 \end{aligned}$$

To make sure we understand how these universal matrices work and their power, let us apply them to a number of specific formulas. Again let us derive Simpson's formula as a check on the method. We evidently want to use $S_3(-1, 0, 1)$. The moments of the operator are

$$\begin{aligned}m_0 &= \int_{-1}^1 1 \cdot dx = 2 \\m_1 &= \int_{-1}^1 x \, dx = 0 \\m_2 &= \int_{-1}^1 x^2 \, dx = 2/3\end{aligned}$$

Therefore we have

$$S_3 m = \begin{pmatrix} 0 & -\frac{1}{2} & \frac{1}{2} \\ 1 & 0 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 2 \\ 0 \\ \frac{2}{3} \end{pmatrix} = \begin{pmatrix} \frac{1}{3} \\ \frac{4}{3} \\ \frac{1}{3} \end{pmatrix} = w$$

which is, of course, the correct answer.

Next consider finding the Simpson's "half-formula"

$$\int_{-1}^0 f(x) \, dx = w_{-1}f(-1) + w_0f(0) + f(1)$$

The moments are

$$\begin{aligned}m_0 &= \int_{-1}^0 1 \cdot dx = 1 \\m_1 &= \int_{-1}^0 x \, dx = -1/2 \\m_2 &= \int_{-1}^0 x^2 \, dx = 1/3\end{aligned}$$

Hence

$$\begin{pmatrix} 0 & -\frac{1}{2} & \frac{1}{2} \\ 1 & 0 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ -\frac{1}{2} \\ \frac{1}{3} \end{pmatrix} = \frac{1}{12} \begin{pmatrix} 5 \\ 8 \\ -1 \end{pmatrix} = w$$

Again, suppose we had tried to find a formula of the form

$$\left. \frac{df}{dx} \right|_{x=0} = w_{-1}f(-1) + w_0f(0) + w_1f(1)$$

The moment vector is

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

and we have

$$\begin{pmatrix} 0 & -\frac{1}{2} & \frac{1}{2} \\ 1 & 0 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix}$$

In the case of

$$\int_{-1}^1 f(x) \sin \frac{\pi}{2} x \, dx$$

the moments were 0 , $8/\pi^2$, and 0 so that

$$\begin{pmatrix} 0 & -\frac{1}{2} & \frac{1}{2} \\ 1 & 0 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 \\ \frac{8}{\pi^2} \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{4}{\pi^2} \\ 0 \\ \frac{4}{\pi^2} \end{pmatrix}$$

Last, suppose we wanted to interpolate at a point x using the same three sample points. This time the moment vector depends on x . We have, after a little thought,

$$\begin{pmatrix} 0 & -\frac{1}{2} & \frac{1}{2} \\ 1 & 0 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 \\ x \\ x^2 \end{pmatrix} = \begin{pmatrix} \frac{x^2 - x}{2} \\ 1 - x^2 \\ \frac{x^2 + x}{2} \end{pmatrix}$$

or more usually

$$f(x) = \frac{x^2 - x}{2} f(-1) + (1 - x^2) f(0) + \frac{x^2 + x}{2} f(1)$$

as the interpolating quadratic, which is clearly the correct answer.

PROBLEMS

Using the universal matrices, do

15.6.1 Prob. 15.4.2.

15.6.2 Prob. 15.4.5. Use S_4 and first move the formula to proper sample points.

15.6.3 Prob. 15.4.4.

15.6.4 Prob. 15.4.3.

15.6.5 Prob. 15.4.1.

15.6.6 Find a quartic interpolation formula through $(-2, -1, 0, 1, 2)$.

15.7 SUMMARY OF THE DIRECT METHOD

The direct method consists of first deciding what *information* (samples) to use and then writing a linear combination of this information with undetermined coefficients. We have chosen in this chapter to confine the discussion to function values at the chosen sample points.

We now impose the conditions that the formula be true for $f(x)$ equal to 1, x , x^2 , ... as far as we can go. This produces the *defining equations*. On one side are the moments of the operator; typically this operator is integration, but we did one problem estimating the derivatives at the origin and another problem estimating an interpolated value at the point x . On the other side of the equation is the Vandermonde matrix of sample points. The inverse of the Vandermonde matrix can be found, if desired, as a combination of sums and products of the sample points in the form of the elementary symmetric functions, each time excluding one sample.

In the special case of equally spaced values the inverse *universal matrices* were tabulated. They show clearly how the method separates the information used (the samples) from the particular formula sought.

The fact that the Vandermonde determinant cannot be zero shows that any formula we want can be found *provided* we can find the moments of the operator we are approximating.

Conventionally we impose the conditions that the formula be true for 1, x , x^2 , x^3 , ..., but a little thought will show that we could take any sequence of polynomials for which the k th polynomial was exactly of degree k (and no lower). The reason is that if the formula is true for all powers lower than k , then putting in either x^k or a polynomial of degree k is the same since automatically all the lower powers of the polynomial will cancel out. Thus there are times when for the convenience of the algebra we will use a sequence of polynomials of degree k . Note that in the case of finding the error we must use a polynomial whose leading coefficient is 1 so that we get the right scale factor.

Further thought shows that *any* set of N linearly independent polynomials could be used in place of 1, x , x^2 , ..., x^{N-1} , and occasionally this freedom is worthwhile.

We have not given long lists of integrals to evaluate; rather we have concentrated on finding formulas. The formulas are meant to be used, but the

process of substituting function values into a formula is so straightforward that it should not require further drill at this stage. But it is important to realize that the formulas are to be used and not merely admired.

15.8 APPENDIX

A number of integrals arise in the course of finding formulas of the form

$$\int_b^a K(x)f(x) dx$$

Among them are:

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-x} x^n dx &= n! \quad n \text{ an integer} \\ \int_{-\infty}^{\infty} e^{-x^2} x^{2n} dx &= \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n} \sqrt{\pi} \quad n \geq 1 \\ \int_{-\infty}^{\infty} e^{-x^2} dx &= \sqrt{\pi} \quad n = 0 \\ - \int_0^1 (\ln x) x^n dx &= \frac{1}{(n+1)^2} \\ \int_0^1 \frac{x^n}{\sqrt{1-x^2}} dx &= \int_0^{\pi/2} \cos^n x dx = \int_0^{\pi/2} \sin^n x dx \\ &= \begin{cases} \frac{1 \cdot 3 \cdot 5 \cdots (n-1)}{2 \cdot 4 \cdots (n)} \frac{\pi}{2} & n = 2, 4, \dots \\ \frac{2 \cdot 4 \cdots (n-1)}{1 \cdot 3 \cdots n} & n = 1, 3, 5, \dots \end{cases} \\ \int_0^{\pi/2} \sin^n x \cos^m x dx &= \frac{\Gamma[(n+1)/2] \Gamma[(m+1)/2]}{2 \Gamma[(m+n+2)/2]} \end{aligned}$$

where $\Gamma(x) = (x-1)\Gamma(x-1)$

$$\Gamma(1/2) = \sqrt{\pi}$$

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ERROR TERMS

16.1 THE NEED OF AN ERROR ESTIMATE

We have adopted an algorithmic approach to finding formulas; we intend to give a rather detailed description of how any of a wide variety of formulas can be found. With this approach we also need a uniform method for finding the error term of the formula, because without an estimate of the error it is difficult to rely on an answer produced by a machine. The error term is also called the *truncation error* or *remainder term*, from the Taylor-series form of derivation (Sec. 15.3). The error is expressed in terms of a high-order derivative, just as in the case of interpolation formulas, and all the faults of higher derivatives (Sec. 14.7) apply; in practice they are both hard to find and likely to be large. The derivatives occur because a polynomial approximation is used in the classical approach.

In this chapter we are concerned with finding the error term, not only for the kinds of formulas we have already examined, but also for those we will take up in the next few chapters. Therefore, we treat the general case, and, as so often happens, the general case is more difficult to carry out than are many trick methods which work in special cases. It is worth accepting this cost in these days of almost infinite knowledge. We also ignore, for the moment, the

question of roundoff in the *use* of the formulas; in the derivations themselves we will watch out for roundoff effects.

Our approach is to first find a formula in terms of an *influence function* which gives the exact expression for the error. But let us be clear about this: if we had the exact error in a useful form, then we would know the exact right answer! When there is an exact expression for an error, it is generally useless, and the more we give up in precision of stating the error, the more useful we can hope it will be. From the influence-function representation of the error we will pass in many cases to a form resembling the mean value theorem with its unknown place of evaluation, θ —the form of the error will be some known constant times a high-order derivative evaluated at some θ in a known interval. From this it is easy to pass to bounds in terms of bounds on the derivative.

When we want to compare formulas, one with another, and both have same-order derivatives in their error terms, it is natural to suppose that the one with the smaller coefficient has the smaller error; but it does not follow. Indeed, we realize that a relatively poor formula will for certain functions give more accurate answer than that of a better formula. Evidently we mean that one formula is better than another formula when measured over some class of functions. Because of the unknown point θ in the error terms we are in fact unable to find the variance over any reasonable ensemble of functions [if $f(x)$ and $-f(x)$ are both in the ensemble of functions with equal weight or probability, then the average would be zero]. Thus we are forced to compare formulas having same-order derivatives by comparing the coefficients of the derivatives, and we will at times make the wrong choice—we hope to be right most of the time.

16.2 THREE BACKGROUND IDEAS

The method of finding the error term for a formula rests on three important results, all of which we have seen in the past, but it is worth reexamining them to be sure we understand what is going on. This is especially necessary because the method, while simple once it is understood, seems to baffle the beginner.

The first result was used in Sec. 4.7. The general expansion of a function in a Taylor series with an integral remainder has the form

$$\begin{aligned} f(x) = & f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots \\ & + \frac{(x-a)^{m-1}}{(m-1)!}f^{(m-1)}(a) + \frac{1}{(m-1)!} \int_a^x f^{(m)}(s)(x-s)^{m-1} ds \end{aligned}$$

If the last term is integrated by parts, we get

$$\begin{aligned}
 u &= (x-s)^{m-1} & du &= -(m-1)(x-s)^{m-2} ds \\
 dv &= f^{(m)}(s) ds & v &= f^{(m-1)}(s)
 \end{aligned}$$

$$\begin{aligned}
 &\frac{1}{(m-1)!} \left[(x-s)^{m-1} f^{(m-1)}(s) \right]_a^x + (m-1) \int_a^x f^{(m-1)}(s) (x-s)^{m-2} ds \\
 &= \frac{1}{(m-1)!} [- (x-a)^{m-1} f^{(m-1)}(a)] + \frac{1}{(m-2)!} \int_a^x f^{(m-1)}(s) (x-s)^{m-2} ds
 \end{aligned}$$

The integrated term cancels the preceding term of the expansion, while the integral is the same as before *except* that the index m is reduced by 1. Thus, a Taylor series with an integral remainder is simply the repeated application of integration by parts (applied in the reverse way, of course).

The second result we need is the weighted mean value theorem for integrals (proved in Sec. 4.7), which states that if $f(x)$ and $g(x)$ are continuous and if $f(x) \geq 0$ in $a \leq x \leq b$, then for some θ in $a < \theta < b$

$$\int_a^b f(x)g(x) dx = g(\theta) \int_a^b f(x) dx$$

The third item is the index m for which x^m first *fails* to satisfy the formula. We began our derivations by making the formula exact for $f(x)$ equal to 1, x , x^2 , ..., etc., and continued until we had enough equations to determine the coefficients in the formula. It sometimes happens that an additional (in principle any number but in practice generally only one) power of x will fit the formula. For example, in Simpson's formula we used 1, x , and x^2 to determine the three coefficients, and we later found that x^3 would also fit the formula exactly. In the Simpson example, $m = 4$.

We need a test to see if by chance the next power of x exactly fits. In the cases we have examined we are given function values at the sample points x_i (which are assumed known), and then all the arithmetic is rational in terms of the moments m_k (Sec. 15.5) and the sample points x_i . In the case of the next chapter, where we are also given derivatives of the function as well as function values, the situation is the same. In Chap. 19 where the x_i are not given in advance, we will have to examine this point further. Thus we assume we can find the value of m which is the lowest power of x that does *not* fit the formula. If we let E_k be the error in the formula when x^k is substituted into it, we have

$$E_0 = E_1 = \cdots = E_{m-1} = 0 \quad E_m \neq 0$$

PROBLEMS

Find m for:

16.2.1 Prob. 15.4.1

16.2.2 Prob. 15.4.3

16.2.3 Prob. 15.4.4

16.3 THE BASIC METHOD APPROACH

The class of formulas we are going to examine has two properties:

- 1 The LHS (left-hand side) is a linear operator.
- 2 The estimation of this operator is made by a linear combination of function values and derivatives of order $\leq m - 2$ where the formula is made exact for $f(x)$ equal to $1, x, x^2, \dots, x^{m-1}$.

It is easiest to think of the operator as integration, but we are not confined to it.

We have the operator equation

$$\text{LHS} = \text{RHS} + \text{remainder} = \text{RHS} + R$$

or

$$R = \text{LHS} - \text{RHS}$$

Supplying the function $f(x)$, we get

$$R[f(x)] = (\text{LHS} - \text{RHS})[f(x)]$$

We next represent $f(x)$ in the Taylor series with an integral remainder

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2}f''(a) + \dots \\ + \frac{(x-a)^{m-1}}{(m-1)!}f^{(m-1)}(a) + \frac{1}{(m-1)!} \int_a^x f^{(m)}(s)(x-s)^{m-1} ds$$

Now let A be the smallest of *all* the values of x that occur any place in the formula, and let B be the largest. In the Taylor expansion we see that (on setting $a = A$)

$$f(x) = P_{m-1}(x) + \frac{1}{(m-1)!} \int_A^x f^{(m)}(s)(x-s)^{m-1} ds$$

where $P_{m-1}(x)$ is a polynomial, in x , of degree $m - 1$.

We made the formula (determined the coefficients and later in Chap. 19 possibly the sample points x_i) so that

$$E_0 = E_1 = \dots = E_{m-1} = 0 \quad E_m \neq 0$$

hence

$$(\text{LHS} - \text{RHS})[P_{m-1}(x)] \equiv 0$$

We have, therefore, only to work with the remainder term of the Taylor series

$$R[f(x)] = (\text{LHS} - \text{RHS}) \left[\frac{1}{(m-1)!} \int_A^x f^{(m)}(s)(x-s)^{m-1} ds \right]$$

16.4 THE INFLUENCE FUNCTION

We now introduce the notation

$$(x-s)_+^j = \begin{cases} 0 & \text{if } s \geq x \\ (x-s)^j & \text{if } s < x \end{cases} \quad j \geq 1$$

For $j = 0$ (see Fig. 16.4.1a):

$$(x-s)_+^0 = \begin{cases} 0 & \text{if } s \geq x \\ 1 & \text{if } s < x \end{cases}$$

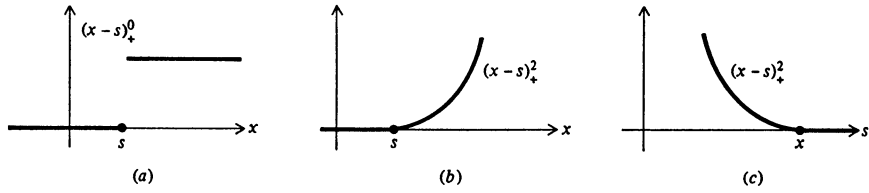


FIGURE 16.4.1

Note that (see Fig. 16.4.1b and c):

$$\frac{d}{dx}(x-s)_+^{m-1} = (m-1)(x-s)_+^{m-2}$$

$$\frac{d^2}{dx^2}(x-s)_+^{m-1} = (m-1)(m-2)(x-s)_+^{m-3}$$

etc., up to derivatives less than¹ than $m-1$. Note also (see Fig. 16.4.2) that

$$\int_a^b (x-s)_+^{m-1} dx = \frac{(b-s)_+^m - (a-s)_+^m}{m} \quad m > 0$$

¹ We can use derivatives of order $m-1$ if we are not frightened of differentiating $(x-s)_+^1$ to get $(x-s)_+^0$.

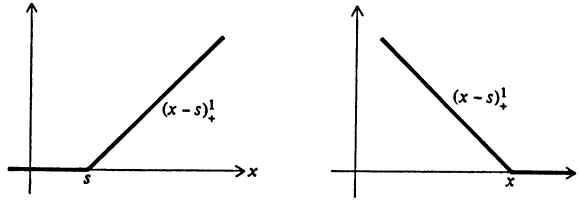


FIGURE 16.4.2

Using this new notation, we may increase the upper limit of the integral to B (B was defined in Sec. 16.3); that is, we have

$$R(f) = (\text{LHS} - \text{RHS}) \left[\frac{1}{(m-1)!} \int_A^B f^{(m)}(s) (x-s)_+^{m-1} ds \right]$$

since we have added nothing in the integrand.

Because everything is linear and the operations in LHS and RHS are on the variable x , not s , we have

$$R(f) = \frac{1}{(m-1)!} \int_A^B f^{(m)}(s) (\text{LHS} - \text{RHS}) [(x-s)_+^{m-1}] ds$$

We set

$$(\text{LHS} - \text{RHS}) \left[\frac{(x-s)_+^{m-1}}{(m-1)!} \right] \equiv G(s)$$

$G(x)$ is called the *influence function*.

Hence

$$R[f(x)] = \int_A^B f^{(m)}(s) G(s) ds$$

Notice that in this expression we have separated the dependence on the formula $G(s)$ from the dependence on the function $f(x)$ we are using.

For the particular function

$$f(x) = x^m$$

we get

$$R[f(x)] = \int_A^B m! G(s) ds \equiv E_m$$

where E_m is, of course, the result of substituting x^m in the formula. Thus we know that

$$\int_A^B G(s) ds = \frac{E_m}{m!} \neq 0$$

An examination of the structure of $G(s)$ shows that $\text{RHS } [f(x)]$ is piecewise a polynomial inside (A, B) and that it is continuous. For many operators the LHS $\{f(x)\}$ is also piecewise continuous.

16.5 WHEN $G(s)$ HAS A CONSTANT SIGN

In most of the important cases it turns out that the influence function $G(s)$ has a constant sign. When this is so, we can use the mean value theorem for integrals and write the remainder, or error term, in the form

$$R = R(f) = \int_A^B f^{(m)}(s)G(s) ds = f^{(m)}(\theta) \int_A^B G(s) ds = \frac{E_m}{m!} f^{(m)}(\theta)$$

where m is the first power of x that does not satisfy the formula.

As a trivial example of this, consider the Taylor series itself. We have

$$G(s) = \frac{(x-s)^{m-1}}{(m-1)!}$$

We then apply the weighted mean value theorem and get

$$\begin{aligned} f(x) - f(a) - (x-a)f'(a) - \frac{(x-a)^2}{2!}f''(a) - \dots \\ - \frac{(x-a)^{m-1}}{(m-1)!}f^{(m-1)}(a) &= \frac{1}{(m-1)!} \int_a^x f^{(m)}(s)(x-s)^{m-1} ds \\ &= \frac{f^{(m)}(\theta)}{(m-1)!} \int_a^x (x-s)^{m-1} ds \\ &= \frac{f^{(m)}(\theta)}{(m-1)!} \frac{(x-a)^m}{m} = \frac{(x-a)^m f^{(m)}(\theta)}{m!} \end{aligned}$$

This form of the Taylor series is widely used in applications.

The central problem is, therefore, how we can determine when $G(s)$ has a constant sign in the interval (A, B) . As a first approach, we examine a number of special cases. Consider first the trapezoid rule

$$R(f) = \int_0^1 f(x) dx - \frac{f(1) + f(0)}{2}$$

which is exact for $f(x)$ equal to 1 and x . Thus $m = 2$, and

$$\begin{aligned} G(s) &= \frac{\text{LHS} - \text{RHS}}{1!} [(x-s)_+^1] \\ 1! G(s) &= \int_0^1 (x-s)_+ dx - \frac{(1-s)_+ + (0-s)_+}{2} \\ &= \frac{(1-s)_+^2 - (0-s)_+^2}{2} - \frac{(1-s)_+ + (0-s)_+}{2} \end{aligned}$$

For $s \geq 1$ (see Fig. 16.5.1)

$$1! G(s) \equiv 0$$

since all the terms vanish. For $1 \geq s \geq 0$

$$1! G(s) = \frac{1}{2} [(1-s)^2 - (1-s)] = \frac{-s(1-s)}{2} \leq 0$$

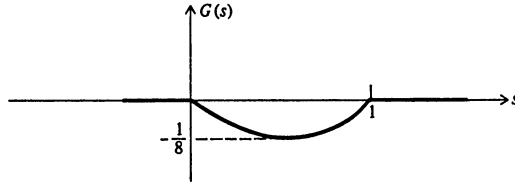


FIGURE 16.5.1

For $0 \geq s$

$$1! G(s) = \frac{1}{2} (1 - 2s + s^2 - s^2 - 1 + s + s) \equiv 0$$

Therefore

$$R(f) = \frac{E_2}{2!} f''(\theta) = -\frac{f''(\theta)}{12}$$

As a second example, consider Simpson's formula, where $m = 4$,

$$\begin{aligned} 3! G(s) &= (\text{LHS} - \text{RHS}) \{(x-s)_+^3\} \\ &= \frac{(1-s)_+^4 - (-1-s)_+^4}{4} - \frac{1}{3} [(-1-s)_+^3 + 4(0-s)_+^3 + (1-s)_+^3] \end{aligned}$$

We have four regions to consider (see Fig. 16.5.2): $s > 1$; $1 \geq s \geq 0$; $0 \geq s \geq -1$; and $-1 \geq s$.

$$s \geq 1: \quad 3!G(s) \equiv 0$$

$$\begin{aligned} 1 \geq s \geq 0: \quad 3!G(s) &= \frac{(1-s)^4}{4} - \frac{(1-s)^3}{3} = (1-s)^3 \left(\frac{1-s}{4} - \frac{1}{3} \right) \\ &= (1-s)^3 \left(\frac{-3s-1}{12} \right) < 0 \end{aligned}$$

$$\begin{aligned} 0 \geq s \geq -1: \quad 3!G(s) &= \frac{(1-s)^4}{4} - \frac{(1-s)^3}{3} - \frac{4(-s)^3}{3} \\ &= \frac{-1 + 6s^2 + 8s^3 + 3s^4}{12} \\ &= (1+s)^3 \left(\frac{3s-1}{12} \right) < 0 \end{aligned}$$

$$-1 \geq s: \quad 3!G(s) \equiv 0 \quad \text{by tedious algebra}$$

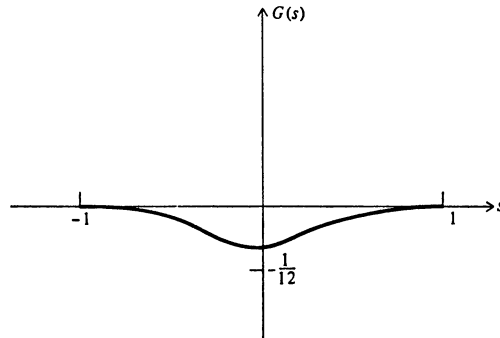


FIGURE 16.5.2

Thus using $E_4 = -\frac{4}{1^5}$ (from Sec. 15.3), we have

$$R(f) = -\frac{4}{15} \frac{f^{(4)}(\theta)}{4!} = -\frac{f^{(4)}(\theta)}{90}$$

as the error of Simpson's formula.

As a third example, consider the formula

$$\int_0^\infty e^{-x} f(x) dx = \frac{f(0) + f(2)}{2}$$

To find m , we try

$$f(x) = 1: \quad \int_0^\infty e^{-x} dx = 1 = \frac{1+1}{2} = 1$$

$$f(x) = x: \quad \int_0^\infty x e^{-x} dx = 1 = \frac{0+2}{2} = 1$$

$$f(x) = x^2: \quad \int_0^\infty x^2 e^{-x} dx = 2 = \frac{0+2^2}{2} = 2$$

$$f(x) = x^3: \quad \int_0^\infty x^3 e^{-x} dx = 3! \neq \frac{0+2^3}{2} = 4$$

Hence $m = 3$, and $E_3 = 3! - 4 = 2$. For $s \geq 0$ (see Fig. 16.5.3)

$$\begin{aligned} 2! G(s) &= (\text{LHS} - \text{RHS})[(x-s)_+^2] \\ &= \int_0^\infty e^{-x} (x-s)_+^2 dx - \frac{(2-s)_+^2 + (0-s)_+^2}{2} \\ &= \int_s^\infty e^{-x} (x-s)_+^2 dx - \frac{(2-s)^2}{2} \\ &= \left[\frac{(x-s)_+^2 e^{-x}}{-1} + \frac{2(x-s)_+ e^{-x}}{-1} + \frac{2e^{-x}}{-1} \right]_s^\infty - \frac{(2-s)_+^2}{2} \\ &= 2e^{-s} - \frac{(2-s)_+^2}{2} \end{aligned}$$

For $s \geq 2$

$$2G(s) = 2e^{-s} > 0$$

For $2 \geq s \geq 0$

$$2G(s) = 2e^{-s} - \frac{(2-s)^2}{2} = \frac{s^2}{2} - \frac{s^3}{3} + \dots$$

requires some examination.

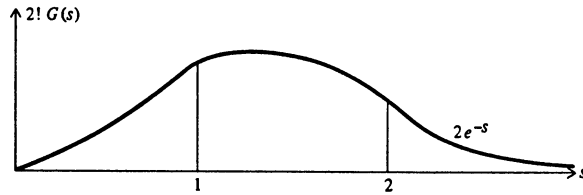


FIGURE 16.5.3

From the series expansion we see that at $x = 0$ the curve is tangent to the s axis and that the second derivative is positive so that the curve is concave upward. A little more effort convinces one that $G(s) \geq 0$ for all s . We had $E_3 = 2$, so that

$$R(f) = \frac{2}{3!} f'''(\theta) = \frac{f'''(\theta)}{3}$$

16.6 THE PRACTICAL EVALUATION OF $G(s)$

The three previous examples show that the method is straightforward, but can be tedious. We need, therefore, to examine the process to see how it can be simplified.

In the first place, it should be clearly recognized that $G(s)$ is piecewise continuous and needs to be evaluated *only once* for a formula. Thus the simple, crude method of using a computer to plot the points with small spacing and looking at the resulting picture can be used. As we did in the third example, a little bit of the calculus can also be used to aid in plotting $G(s)$.

There are further items to be noted. First, outside the range of all the function values $G(s) = 0$. This follows easily from noting that if s is greater than every sample point and value used in the operator, then all the values will be zero. At the other end of the range of s , all the terms will be there, and since the formula is true for a polynomial in x of degree $m - 1$ and in that range we have $(x - s)^{m-1}$ which is a polynomial of degree $m - 1$, we know that $G(s)$ will vanish identically.

We notice that as we progress from right to left, taking more and more terms, the algebra becomes more difficult. This can be easily remedied by using a second notation

$$(x - s)_-^j = \begin{cases} 0 & s \leq x \\ (x - s)^j & s > x \end{cases} \quad j \geq 0$$

whose meaning is obvious. Note that

$$(x - s)_-^j + (x - s)_+^j \equiv (x - s)^j$$

Now, since

$$(\text{LHS} - \text{RHS})[(x - s)^j] \equiv 0 \quad j < m$$

$$(\text{LHS} - \text{RHS}) \left[\frac{(x - s)_+^{m-1}}{(m-1)!} \right] = -(\text{LHS} - \text{RHS}) \left[\frac{(x - s)_-^{m-1}}{(m-1)!} \right]$$

we can start from the other end of the interval and use fewer terms for a while.

When the formula is symmetric, as Simpson's formula was, then $G(s)$ is also symmetric [notice how replacing s by $-s$ transforms one expression for $G(s)$ into

the other] so that only half the range need be examined. The formula may be symmetric about a point $x \neq 0$, and $G(s)$ will also be symmetric about that point.

We need, in the method, the value of E_m . We can find this directly from Chap. 15 from knowing the sample points, without finding the weights. All we need to do is compute, using $\pi(x) = (x - x_1)(x - x_2) \cdots (x - x_n)$, (or $x\pi(x)$),

$$R[\pi(x)] = (\text{LHS} - \text{RHS})[\pi(x)] = \text{LHS}[\pi(x)] = E_m$$

since $\pi(x)$ is zero at each sample point on the right-hand side. Thus *before* we get involved in the details of the formula and in finding $G(s)$, we can find the error [if $G(s)$ turns out to be of constant sign].

PROBLEMS

16.6.1 Show that the three-eighths rule is valid.

$$\int_0^3 f(x) dx = \frac{3}{8} [f(0) + 3f(1) + 3f(2) + f(3)] - \frac{3}{80} f^{(4)}(\theta)$$

16.6.2 Show that the midpoint and trapezoid rules have errors which are in the ratio of $-1/2$ to each other.

16.6.3 Show that

$$\int_{-1}^1 f(x) dx = \frac{1}{4} \left[3f\left(-\frac{2}{3}\right) + 2f(0) + 3f\left(\frac{2}{3}\right) \right] + \frac{7}{1,620} f^{(4)}(\theta)$$

16.6.4 Find the error term of Prob. 15.4.3.

16.6.5 Find the error term of Prob. 15.4.4.

16.6.6 Find the error term of Prob. 15.4.5.

16.7 WHEN $G(s)$ IS NOT OF CONSTANT SIGN

When $G(s)$ is not of constant sign, we *cannot* express the error as we did in the previous section. Indeed, for some functions the error cannot be so expressed. To show this, suppose that $G(s)$ is positive except for some small interval and that the integral of $G(s)$ is positive (see Fig. 16.7.1). Consider now a function $f(x)$ whose m th derivative is positive and continuous but which is small outside and large inside the interval in which $G(s)$ is negative. Thus the integral

$$\int_A^B f^{(m)}(s)G(s) ds$$

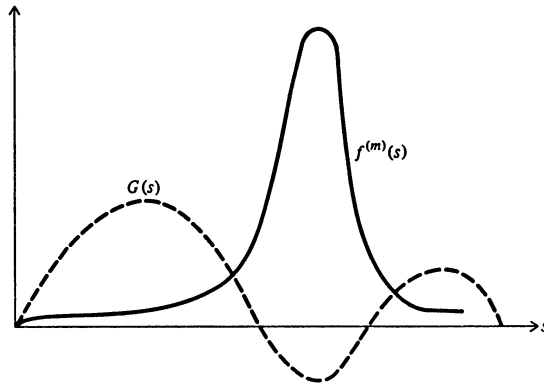


FIGURE 16.7.1

will be negative, but the product

$$f^{(m)}(s) \int_A^B G(s) ds$$

will be positive. Hence there can be no θ in such a case, and we have proved

Theorem 16.7.1 If $G(s)$ is not of constant sign, then there are functions which cannot have their error terms written in the form

$$\frac{E_m}{m!} f^{(m)}(\theta) = f^{(m)}(\theta) \int_A^B G(s) ds$$

The second point to observe is that when $G(s)$ changes sign, we can still get a bound on the error. The bound follows readily by taking the absolute value of $G(s)$:

$$\int_A^B f^{(m)}(s) |G(s)| ds = f^{(m)}(\theta) \int_A^B |G(s)| ds$$

Hence

$$\left| \int_A^B f^{(m)}(s) G(s) ds \right| \leq \max\{|f^{(m)}(x)|\} \int_A^B |G(s)| ds$$

where the max is over all x ($A \leq x \leq B$). The computation of the integral

$$\int_A^B |G(s)| ds$$

can be done analytically or estimated numerically since, again, it need be done only once for any particular formula.

The third observation is that when the sign of $G(s)$ changes, then we can still find an error term. As an illustration, consider the function $G(s)$ as shown in Fig. 16.7.2. The point $x = C$ has been chosen so that the two shaded areas are equal in size; that is,

$$\int_C^B G(s) ds = 0$$

We have

$$\begin{aligned} R = R(f) &= \int_A^B f^{(m)}(s)G(s) ds \\ &= \int_A^C + \int_C^B \end{aligned}$$

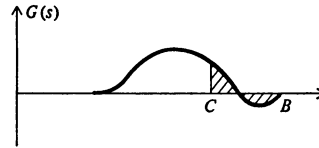


FIGURE 16.7.2

We integrate the second term by parts and set

$$\int_C^s G(s) ds = H(s) \quad \text{Note: } H(B) = H(C) = 0.$$

Then

$$\begin{aligned} R &= \int_A^C f^{(m)}(s)G(s) ds + f^{(m)}(s)H(s) \Big|_C^B - \int_C^B f^{(m+1)}(s)H(s) ds \\ &= \int_A^C f^{(m)}(s)G(s) ds - \int_C^B f^{(m+1)}(s)H(s) ds \end{aligned}$$

We can now apply the mean value theorem to each piece.

$$R = f^{(m)}(\theta_1) \int_A^C G(s) ds - f^{(m+1)}(\theta_2) \int_C^B H(s) ds$$

If $G(s)$ is as shown in Fig. 16.7.3, we could simply set

$$H(s) = \int_A^s G(s) ds$$

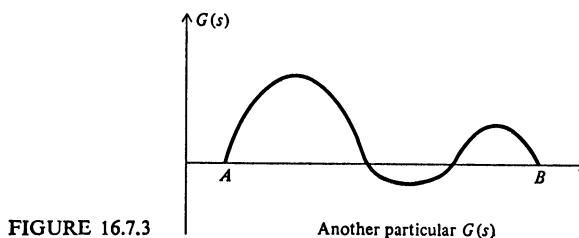


FIGURE 16.7.3

and integrate this by parts

$$\begin{aligned} R(f) &= f^{(m)}(s)H(s) \Big|_A^B - \int_A^B f^{(m+1)}(s)H(s) ds \\ &= f^{(m)}(B)H(B) - f^{(m+1)}(\theta) \int_A^B H(s) ds \end{aligned}$$

If we put $f(x) = x^{m+1}$ in this formula, we get

$$R(f) \equiv E_{m+1} = (m+1)! BH(B) - (m+1)! \int_A^B H(s) ds$$

so that

$$\int_A^B H(s) ds = \frac{E_{m+1}}{(m+1)!} - B \frac{E_m}{m!}$$

Therefore we have in general

$$R(f) = \frac{f^{(m)}(B)E_m}{m!} - \frac{f^{(m+1)}(\theta)}{(m+1)!} [E_{m+1} - (m+1)BE_m]$$

We shall not analyze the general case, since the results are not used in the text, but shall merely observe that such forms can be useful at times.

16.8 THE FLAW IN THE TAYLOR-SERIES APPROACH

Since there exists an error term from the Taylor-series of the form¹

$$\begin{aligned} f(a+x) &= f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \cdots \\ &\quad + \frac{(x-a)^{m-1}}{(m-1)!} f^{(m-1)}(a) + \frac{(x-a)^m f^{(m)}(a+\theta x)}{m!} \end{aligned}$$

¹ See Sec. 16.5 for this Lagrange form.

it is natural to ask why we cannot use the Taylor-series approach (second method, Sec. 15.3) to obtain the error of the formula. Let us suppose that we do use the Taylor-series approach. The coefficients of the formula have been determined to make the powers of x cancel on both sides up to the m th power, which is assumed not to cancel. Let us fix the x_i values in our mind. We therefore get a series of terms in $f^{(m)}(a + \theta_i x_i)$ for various θ_i

$$\alpha_1 f^{(m)}(a + \theta_1 x_1) + \alpha_2 f^{(m)}(a + \theta_2 x_2) + \cdots + \alpha_k f^{(m)}(a + \theta_k x_k)$$

where, of course, the θ_i depend on x_i , $\theta_i = \theta_i(x_i)$. We now want to replace this expression by

$$(\alpha_1 + \alpha_2 + \cdots + \alpha_k) f^{(m)}(a + \theta x)$$

for some θ (and we expect some α_i to be positive and some to be negative and hence some cancellation). Under what circumstances does there exist such a θ ?

We know from the influence-function approach that if $G(s)$ is of constant sign, then there is such a θ and that if $G(s)$ is not of constant sign, then there are some functions $f(x)$ for which there is no such θ . In this indirect manner we can answer the question. There seems to be no direct way of finding the answer, but if one were found, then it might provide an alternative, and perhaps better, approach to the theory of the error term.

16.9 A CASE STUDY

The idea that $G(s)$ does not change sign except in very special circumstances is apt to emerge from the study of classical formulas where it is always of constant sign. The following example is therefore illuminating.

Consider the family of integration formulas

$$\int_{-1}^1 f(x) dx = w_{-1}f(-t) + w_0f(0) + w_1f(t)$$

where t is a parameter (see Fig. 16.9.1). This family includes a number of interesting cases. We are sampling the integrand at the points $-t$, 0 , and t and are weighting the samples by w_{-1} , w_0 , and w_1 . We will only consider $0 \leq t \leq 1$ and ignore $t > 1$.

By symmetry it is easy to see that $w_{-1} = w_1$. The defining equations are, therefore,

$$\begin{aligned} f \equiv 1: & \quad 2 = 2w_1 + w_0 \\ f \equiv x^2: & \quad 2/3 = 2w_1 t^2 \\ f \equiv x^4: & \quad 2/5 = 2w_1 t^4 + E_4 \end{aligned}$$

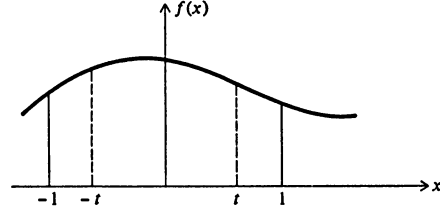


FIGURE 16.9.1

From the second equation we have

$$w_1 = \frac{1}{3t^2}$$

and hence from the first

$$w_0 = 2\left(1 - \frac{1}{3t^2}\right)$$

and from the third we get

$$E_4 = 2\left(\frac{1}{5} - \frac{t^2}{3}\right)$$

We need to examine the influence function $G(s)$ (see Fig. 16.9.2). Again by symmetry we need only examine the two intervals $1 \geq s \geq t$ and $t \geq s \geq 0$. We have, since $m = 4$,

$$\begin{aligned} 3! G(s) = & \frac{(1-s)_+^4 - (-1-s)_+^4}{4} - \frac{1}{3t^2} [(t-s)_+^3 - (-t-s)_+^3] \\ & - 2\left(1 - \frac{1}{3t^2}\right)(-s)_+^3 \end{aligned}$$

For $1 \geq s \geq t$

$$3! G(s) = \frac{(1-s)^4}{4} \geq 0$$

For $t \geq s \geq 0$

$$3! G(s) = \frac{(1-s)^4}{4} - \frac{(t-s)^3}{3t^2}$$

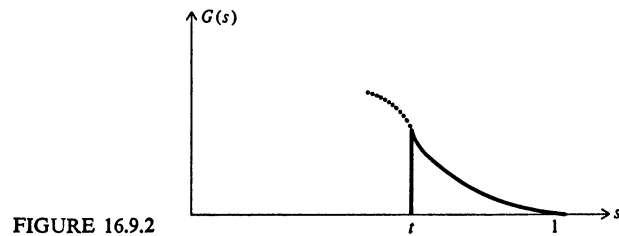


FIGURE 16.9.2

Can this change sign? To find out, we evaluate it at $s = 0$

$$3!G(0) = 1/4 - \frac{t}{3}$$

Hence for $t = 3/4$, $G(0) = 0$; for $t > 3/4$, $G(0) < 0$, and the influence function $G(s)$ changes sign (see Fig. 16.9.3). Thus

for $3/4 < t < 1$ $G(s)$ changes sign

for $0 \leq t \leq 3/4$ and $t = 1$ $G(s)$ has constant sign

The case $t = 1$ requires special investigation and is Simpson's formula

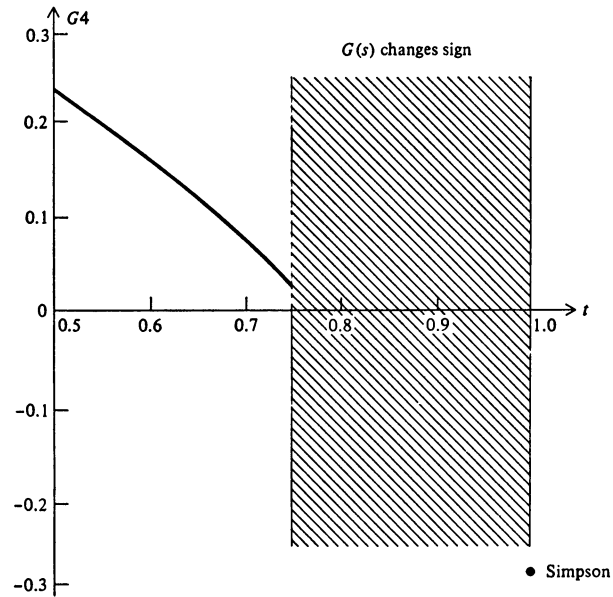


FIGURE 16.9.3

If $G(s)$ does not change sign, then the error is

$$\frac{E_4 f^{(4)}(\theta)}{4!} = \frac{1}{12} \left(\frac{1}{5} - \frac{t^2}{3} \right) f^{(4)}(\theta)$$

The case $E_4 = 0$, which is

$$t = \sqrt{3/5} = 0.7745967 \dots$$

immediately attracts attention. It falls in the region $3/4 < t < 1$ of sign change. And this is typical; in the region of sign change there is apt to be a place where the total area under $G(s)$ is exactly zero, meaning that $E_m = 0$, and the formula will have extra accuracy [and a different $G(s)$ will have to be investigated using the appropriate m , in this particular case, two higher than before].

Table 16.9 shows a number of special cases of interest in the family of formulas. A study of this table shows how the error changes as we change where we sample the integrand. The beginner should study families of formulas to get a feel for the effects of where samples are taken and how the weights change. From this can come a feeling of what can be expected from some new formula before it is investigated.

Table 16.9

t	w_0	w_1	E_4	Remarks
$1/2$	$-2/3$	$4/3$	$7/30 = 0.233$	Bad roundoff properties
$1/\sqrt{3}$	0	1	$8/45 = 0.1778$	2 point Gauss (Chap. 19)
$2/3$	$2/4$	$3/4$	$14/135 = 0.1037$	Better than twice as accurate as Simpson's formula
$1/\sqrt{2}$	$2/3$	$2/3$	$1/15 = 0.0667$	Chebyshev (Chap. 19)
$3/4$	$22/27$	$16/27$	$1/40 = 0.0250$	Better than 10 times as accurate as Simpson's formula
$\sqrt{3/5}$	$8/9$	$5/9$	$0 = 0.0000$	3 point Gauss (Chap. 19)
1	$4/3$	$1/3$	$-4/15 = -0.2667$	Simpson's formula

17.1 INTRODUCTION

In the past three chapters we have studied, respectively, interpolation, finding formulas, and finding their error terms, using only function values. In this chapter we will repeat the work, this time using values of the function and of one or more derivatives.

Why should we care about using derivatives? Experimentally they are hard to come by, but when the function is given in a mathematical form, then they are easy to get. This remark contradicts one's initial impression from having taken a calculus course where differentiation tends to spread out the formula over more and more paper as the derivatives get higher and higher. This spreading out is indeed true, but it is not really relevant to modern computing. When an examination is made of how machine time is spent, it turns out that the evaluations of square roots, exponentials, logarithms, arctangents, and sines and cosines are done by subroutines which usually take more time than 50 arithmetic operations. But differentiation *does not* introduce any new radicals, exponentials, logarithms, or arctangents, and at worst a sine goes into a cosine and a cosine goes into a sine of the same angle. Thus, while the formula for the derivative may appear much

longer when written out on paper, it will usually take much less time to evaluate *provided* the costly parts of the function evaluation have been saved to be used again. As we shall see, the value of a derivative at a point is worth almost as much as another function value at some new point.

We can expect, therefore, that when pressed for machine time, it will be worth considering formulas which use one or more derivatives as an alternative to using more function values. In this chapter, we show how to derive such formulas.

17.2 HERMITE¹ INTERPOLATION

Hermite interpolation uses both the function value and the value of the first derivative at each point. Thus the interpolating polynomial is tangent to the function at the sample points, and Hermite interpolation is sometimes known as “osculating interpolation.” The first approach to Hermite interpolation will be similar to our approach to Lagrange interpolation (Sec. 14.5). In place of the sampling polynomials $\pi_i(x)$ of the Lagrange formula we need to find two such families of functions, the $\sigma_i(x)$, which are zero at all the data except the i th function value, and the $\tau_i(x)$, which are zero at all the data except the i th first-derivative value. In mathematical symbols

$$\begin{aligned} \sigma_i(x_j) &= \begin{cases} 0 & i \neq j \\ \neq 0 & i = j \end{cases} & \tau_i(x_j) &= 0 \quad \text{for all } j \\ \sigma'_i(x_j) &= 0 \quad \text{for all } j & \tau'_i(x_j) &= \begin{cases} 0 & i \neq j \\ \neq 0 & i = j \end{cases} \end{aligned}$$

We begin by constructing the $\tau_i(x)$. Evidently, for all the sample points, except for x_i , we have double zeros, and at x_i we have a single zero. Therefore, our sampling polynomial is

$$\tau_i(x) = \pi_i^2(x)(x - x_i)$$

where as before

$$\pi_i(x) = (x - x_1)(x - x_2) \cdots (x - x_{i-1})(x - x_{i+1}) \cdots (x - x_{N+1})$$

The degree of $\tau_i(x)$ in x is $2N + 1$, as it should be.

The $\sigma_i(x)$ are less obvious, so we assume the reasonable form

$$\sigma_i(x) = \pi_i^2(x)(x + b)$$

where $x_i + b \neq 0$. Differentiating, we get

$$\sigma'_i(x) = 2\pi_i(x)\pi'_i(x)(x + b) + \pi_i^2(x)$$

¹ Charles Hermite (1822–1901).

But

$$\sigma'_i(x_i) = 0$$

so we have

$$\begin{aligned} \frac{\sigma'_i(x_i)}{\sigma_i(x_i)} &= \frac{2\pi_i(x_i)\pi'_i(x_i)(x_i + b) + \pi_i^2(x_i)}{\pi_i^2(x_i)(x_i + b)} = 0 \\ &= 2 \frac{\pi'_i(x_i)}{\pi_i(x_i)} + \frac{1}{x_i + b} = 0 \\ &= 2 \sum_{j \neq i} \frac{1}{x_i - x_j} + \frac{1}{x_i + b} = 0 \end{aligned}$$

which determines b .

Again, the important part is *how* we found the $\sigma_i(x)$ and $\tau_i(x)$, not the result.

Using these functions makes it easy to see that the Hermite interpolating polynomial is

$$y(x) = \sum_{i=1}^N \left[\frac{\sigma_i(x)}{\sigma_i(x_i)} y_i + \frac{\tau_i(x)}{\tau'_i(x_i)} y'_i \right]$$

To find the error term, we proceed *exactly* as we did in Sec. 14.6. We know that the difference between the function $y(x)$ and the approximating polynomial $P_{2N+1}(x)$ has double zeros at each x_i , and so we write the difference as

$$y(x) = P_{2N+1}(x) + \pi^2(x)K(x)$$

where, as usual,

$$\pi(x) = (x - x_1)(x - x_2) \cdots (x - x_{N+1})$$

We now pick an arbitrary $x^* \neq x_j$ (all j) such that

$$y(x^*) = P_{2N+1}(x^*) + \pi^2(x^*)K(x^*)$$

and then consider the expression

$$\phi(x) = y(x) - P_{2N+1}(x) - \pi^2(x)K(x^*)$$

This has double zeros at each x_i and a simple zero at x^* . Using a simple extension of the mean value theorem, we have that the first derivative has at least $2N + 2$ zeros, the second derivative $2N + 1$ zeros, ..., the $(2N + 2)$ nd derivative at least 1 zero, say $\bar{x}$. Hence

$$y^{(2N+2)}(\bar{x}) = (2N + 2)! K(x^*)$$

and putting this in the original expression for $\phi(x)$, we get

$$y(x^*) = P_{2N+1}(x^*) + \frac{\pi^2(x^*)}{(2N + 2)!} y^{(2N+2)}(\bar{x})$$

But, since x^* was arbitrary, we may write x^* as x , and we have the formula

$$y(x) = P_{2N+1}(x) + \frac{\pi^2(x)}{(2N+2)!} y^{(2N+2)}(\bar{x})$$

for the error of Hermite interpolation.

PROBLEMS

17.2.1 Construct the Hermite interpolating polynomial on $x = 0$ and $x = 1$.

17.2.2 Apply Hermite interpolation to

$$y = \sin x \quad \text{using} \quad \begin{array}{ll} \sin 0^\circ = 0.0000 & \cos 0^\circ = 1.0000 \\ \sin 30^\circ = 0.5000 & \cos 30^\circ = 0.8660 \end{array}$$

(beware of radians and degrees!) and compute $\sin 15^\circ$.

17.2.3 Using 0° , 30° , and 60° , by the Hermite interpolation find $\sin 15^\circ$.

17.2.4 Carry out the theory for y , y' , and y'' .

17.3 THE DIRECT METHOD

In the direct method for finding the Hermite interpolating polynomial of degree $2N+1$ we simply write down the form

$$P_{2N+1}(x) = a_0 + a_1x + \cdots + a_{2N+1}x^{2N+1}$$

and impose the conditions on the polynomial to get the defining equations (where we have put the derivative values after the function values)

$$\begin{array}{llll} P_{2N+1}(x_1) & = & a_0 + a_1x_1 & + a_2x_1^2 + \cdots + a_{2N+1}x_1^{2N+1} \\ P_{2N+1}(x_2) & = & a_0 + a_1x_2 & + a_2x_2^2 + \cdots + a_{2N+1}x_2^{2N+1} \\ \cdot & \cdot & \cdot & \cdot \\ P_{2N+1}(x_{N+1}) & = & a_0 + a_1x_{N+1} & + a_2x_{N+1}^2 + \cdots + a_{2N+1}x_{N+1}^{2N+1} \\ P'_{2N+1}(x_1) & = & a_1 & + 2a_2x_1 + \cdots + (2N+1)a_{2N+1}x_1^{2N} \\ \cdot & \cdot & \cdot & \cdot \\ P'_{2N+1}(x_{N+1}) & = & a_1 & + 2a_2x_{N+1} + \cdots + (2N+1)a_{2N+1}x_{N+1}^{2N} \end{array}$$

We will later show that the determinant of these equations is not zero, and hence we can solve directly for the interpolating polynomial.

We recognize that if we were to integrate this polynomial, we would get a linear combination of the data

$$\int_a^b P_{2N+1}(x) dx = \sum_{k=1}^{N+1} [w_k f(x_k) + w'_k f'(x_k)]$$

Knowing the form, let us apply the direct method for finding the formula for integration that uses both the function values and the values of the first derivative at each point. Using $f(x) = x^j$, we get the defining equations

$$x^j: \quad m_j = \sum_{k=1}^{N+1} [w_k x_k^j + w'_k j x_k^{j-1}] \quad j = 0, 1, \dots, 2N+1$$

The matrix of these equations is exactly the *transpose* of the earlier set of equations used to find the Hermite interpolating polynomial.

We shall solve these equations by what amounts to finding the inverse determinant. If we write the $\sigma_i(x)$ as polynomials, we get

$$\sigma_i(x) = s_{i,0} + s_{i,1}x + \dots + s_{i,2N+1}x^{2N+1}$$

Now if for each j we multiply the j th row by $s_{i,j}$ and add all the resulting equations, we get, after some careful thought about what the $\sigma_i(x)$ actually are [note that $\sigma_i(x_j) = 0$ for $i \neq j$ and $\sigma'_i(x_j) = 0$ for all j],

$$\sum_{j=0}^{2N+1} s_{i,j} m_j = \sum_{k=1}^{N+1} [w_k \sigma_i(x_k) + w'_k \sigma'_i(x_k)] = w_i \sigma_i(x_i)$$

Similarly, expanding the

$$\tau_i(x) = t_{i,0} + t_{i,1}x + \dots + t_{i,2N+1}x^{2N+1}$$

we are led to

$$\sum_{j=0}^{2N+1} t_{i,j} m_j = w'_i \tau'_i(x_i)$$

Thus we have, for all practical purposes, the inverse of the corresponding determinant. Since we have found the inverse, the determinant could not have been zero.

The method seems a bit magical so we will illustrate it in the next section by applying it to two particular cases. Again, it is the method more than the result that we care about.

17.4 THE HERMITE UNIVERSAL MATRICES

The process of using the sampling polynomials is so useful in the practical work of finding formulas, and the inverse matrices are so useful, that it is worth developing the cases of two and three equally spaced data points.

In the case of two points, we use $x = -1/2$ and $x = 1/2$ as the sample points. For any linear operator $L[f(x)]$ we have the defining equations.

$$\begin{aligned} L(1) &= m_0 = w_1 + w_2 \\ L(x) &= m_1 = w_1(-1/2) + w_2(1/2) + w'_1 + w'_2 \\ L(x^2) &= m_2 = w_1(-1/2)^2 + w_2(1/2)^2 + 2[w'_1(-1/2) + w'_2(1/2)] \\ L(x^3) &= m_3 = w_1(-1/2)^3 + w_2(1/2)^3 + 3[w'_1(-1/2)^2 + w'_2(1/2)^2] \end{aligned}$$

From symmetry it is obvious that the two $\sigma_i(x)$ are related, $\sigma_{1/2}(x) = \sigma_{-1/2}(-x)$

$$\begin{aligned} \sigma_{-1/2} &= (x - 1/2)^2(x + b) \\ \sigma'_{-1/2} &= 2(x - 1/2)(x + b) + (x - 1/2)^2 \end{aligned}$$

We know $\sigma'_{-1/2}(-1/2) = 0$, and so we have

$$\begin{aligned} -2(-1/2 + b) + 1 &= 0 \\ b &= 1 \end{aligned}$$

and therefore

$$\sigma_{-1/2}(-1/2) = 1/2$$

Hence

$$\frac{\sigma_{-1/2}(x)}{\sigma_{-1/2}(-1/2)} = 2x^3 - \frac{3}{2}x + \frac{1}{2}$$

gives the elements of the inverse matrix (namely, $1/2$, $-3/2$, 0 , and 2). Correspondingly, we have for $\tau_{1/2}(x) = -\tau_{-1/2}(-x)$

$$\begin{aligned} \tau_{-1/2} &= (x - \frac{1}{2})^2(x + \frac{1}{2}) = x^3 - \frac{1}{2}x^2 - \frac{1}{4}x + \frac{1}{8} \\ \tau'_{-1/2} &= 2(x - \frac{1}{2})(x + \frac{1}{2}) + (x - \frac{1}{2})^2 \\ \tau'_{-1/2}(-1/2) &= 1 \\ \frac{\tau_{-1/2}(x)}{\tau_{-1/2}(-1/2)} &= x^3 - \frac{1}{2}x^2 - \frac{1}{4}x + \frac{1}{8} \end{aligned}$$

Using these coefficients, we have the inverse matrix on the points $x = -1/2$ and $x = 1/2$

$$\begin{array}{c} \frac{\sigma_{-1/2}(x)}{\sigma_{-1/2}(-1/2)} \\ \frac{\sigma_{1/2}(x)}{\sigma_{1/2}(1/2)} \\ \frac{\tau_{-1/2}(x)}{\tau_{-1/2}(-1/2)} \\ \frac{\tau_{1/2}(x)}{\tau_{1/2}(1/2)} \end{array} \begin{pmatrix} 1 & x & x^2 & x^3 \\ 1/2 & -3/2 & 0 & 2 \\ 1/2 & 3/2 & 0 & -2 \\ 1/8 & -1/4 & -1/2 & 1 \\ -1/8 & -1/4 & 1/2 & 1 \end{pmatrix}$$

For three points x is equal to -1 , 0 , and 1 and we have for the inverse

$$\frac{1}{4} \begin{pmatrix} 0 & 0 & 4 & -5 & -2 & 3 \\ 4 & 0 & -8 & 0 & 4 & 0 \\ 0 & 0 & 4 & 5 & -2 & -3 \\ 0 & 0 & 1 & -1 & -1 & 1 \\ 0 & 4 & 0 & -8 & 0 & 4 \\ 0 & 0 & -1 & -1 & 1 & 1 \end{pmatrix}$$

We derive only one row, say the fifth, since the others are similar.

$$\tau_0(x) = (x-1)^2(x+1)^2x = 0 + x + 0x^2 - 2x^3 + 0x^4 + x^5$$

$$\tau'_0(x) = 1 - 6x^2 + 5x^4$$

$$\tau'_0(0) = 1$$

Hence the coefficients are $0, 1, 0, -2, 0$, and 1 , and the $1/4$ in front adjusts the entries correctly.

Higher-order cases of the inverse matrix can be done but are not likely to be useful.

PROBLEMS

$$17.4.1 \quad \text{Find } \int_0^{2h} f(x) dx = \frac{h}{15} [7f(0) + 16f(h) + 7f(2h)] \\ + \frac{h^2}{15} [f'(0) - f'(2h)] + \frac{h^7 f^{(6)} \theta}{4,725}$$

$$17.4.2 \quad \text{Using both matrices, find } -\int_0^1 f(x) \ln x dx.$$

$$17.4.3 \quad \text{Find } \int_0^1 f(x) dx \text{ for 2 points.}$$

$$17.4.4 \quad \text{Find } \int_0^1 f(x) dx \text{ using } x = 1/4 \text{ and } x = 3/4 \text{ as the sample points for } f(x) \text{ and } f'(x).$$

17.5 SOME EXAMPLES

We propose to find some typical formulas as illustrations of the method of using derivatives as well as function values.

EXAMPLE 17.1 Find

$$\int_0^1 f(x) dx = w_0 f(1/2) + w'_0 f'(1/2)$$

$$\text{if } f(x) = 1 \quad \text{then} \quad 1 = w_0$$

$$\text{if } f(x) = x \quad \text{then} \quad 1/2 = w_0(1/2) + w'_0, w'_0 = 0$$

To get E_2 we use $f(x) = x^2$

$$1/3 = w_0(1/2)^2 + 2w'_0(1/2) + E_2$$

$$E_2 = 1/3 - 1/4 = 1/12$$

The $G(s)$ is ($m = 2$)

$$1! G(s) = \frac{(1-s)_+^2 - (0-s)_+^2}{2} - (1/2-s)_+$$

For $1 \geq s \geq 1/2$

$$G(s) = \frac{(1-s)^2}{2} \geq 0$$

By symmetry, or further algebra, we conclude that $G(s) \geq 0$ ($0 \leq s \leq 1$), and we have the midpoint formula (again)

$$\int_0^1 f(x) dx = f(1/2) + \frac{1}{24} f''(\theta) \quad ////$$

EXAMPLE 17.2 Find

$$\int_0^1 f(x) dx = w_0 f(0) + w_1 f(1) + w'_0 f'(0) + w'_1 f'(1)$$

We can use the inverse in Sec. 17.4 as a check. The defining equations are

$$1 = w_0 + w_1$$

$$1/2 = w_0 + w_1 + w'_0 + w'_1$$

$$1/3 = w_0 + w_1 + 2w'_1$$

$$1/4 = w_0 + w_1 + 3w'_1$$

$$1/5 = w_0 + w_1 + 4w'_1 + E_4$$

The fourth equation minus the third gives

$$1/4 - 1/3 = w'_1 = -1/12$$

and the rest is easy, $w_1 = 1/2$, $w_0 = 1/2$, $w'_0 = 1/12$, $E_4 = 1/30$

$$\int_0^1 f(x) dx = \frac{1}{12} [6f(0) + 6f(1) + f'(0) - f'(1)] + \frac{f^{(4)}(\theta)}{720}$$

To check $G(s)$, we have

$$3!G(s) = (\text{LHS} - \text{RHS})[(x-s)_+^3]$$

and for $1 \geq s \geq 0$

$$\begin{aligned} 3!G(s) &= \frac{(1-s)_+^4}{4} - \frac{1}{12} [6(1-s)_+^3 - 3(1-s)_+^2] \\ &= \frac{(1-s)^2}{4} [1 - 2s + s^2 - 2 + 2s + 1] = \left[\frac{s(1-s)}{2} \right]^2 \geq 0 \quad //// \end{aligned}$$

EXAMPLE 17.3 Find

$$\begin{aligned} \int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx &= a_{-1}f(-1) + a_0f(0) + a_1f(1) \\ &\quad + b_{-1}f'(-1) + b_0f'(0) + b_1f'(1) \end{aligned}$$

By now we can begin to recognize symmetry and realize that if in this problem we set

$$\begin{aligned} a_{-1} &= a_1 \\ b_{-1} &= -b_1 \quad b_0 = 0 \end{aligned}$$

then the formula will be true for any odd power of x . Thus we only need to use $f(x)$ equal to 1, x^2 , and x^4 :

$$\begin{aligned} f(x) = 1 : \quad \pi &= 2a_1 + a_0 \\ f(x) = x^2 : \quad \frac{\pi}{2} &= 2a_1 + 2(2b_1) \\ f(x) = x^4 : \quad \frac{3\pi}{8} &= 2a_1 + 4(2b_1) \end{aligned}$$

from which it is easy to find

$$b_1 = \frac{-\pi}{32} \quad a_1 = \frac{5\pi}{16} \quad a_0 = \frac{6\pi}{16}$$

Using x^6

$$\begin{aligned} \frac{5\pi}{16} &= 2a_1 + 6(2b_1) + E_6 \\ E_6 &= \frac{\pi}{16} \end{aligned}$$

Thus we have

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = \frac{\pi}{32} [10f(-1) + 12f(0) + 10f(1) + f'(-1) - f'(1)] + \frac{\pi}{16 \cdot 6!} f^{(6)}(\theta)$$

We can check the algebra by using the universal matrix

$$\frac{1}{4} \begin{pmatrix} 0 & 0 & 4 & -5 & -2 & 3 \\ 4 & 0 & -8 & 0 & 4 & 0 \\ 0 & 0 & 4 & 5 & -2 & -3 \\ 0 & 0 & 1 & -1 & -1 & 1 \\ 0 & 4 & 0 & -8 & 0 & 4 \\ 0 & 0 & -1 & -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \pi \\ 0 \\ \pi/2 \\ 0 \\ 3\pi/8 \\ 0 \end{pmatrix} = \frac{\pi}{32} \begin{pmatrix} 10 \\ 12 \\ 10 \\ 1 \\ 0 \\ -1 \end{pmatrix}$$

We need, still, to check $G(s)$ and due to symmetry we have only to examine

$$1 \geq s \geq 0$$

We leave the details to the reader.

////

PROBLEMS

17.5.1 Derive

$$\int_0^\infty e^{-x} f(x) dx = w_0 f(0) + w_2 f(2) + w'_0 f'(0) + w'_2 f'(2)$$

$$\text{Ans. } w_0 = 1, w_2 = 0, w'_0 = w'_2 = 1/2$$

17.5.2 Derive

$$\int_0^\infty e^{-x} f(x) dx = w_0 f(0) + w_2 f(2) + w_4 f'(4) + w'_0 f'(0) + w'_2 f'(2) + w'_4 f'(4)$$

17.5.3 Derive

$$-\int_0^1 (\ln x) f(x) dx$$

using x equal to 0, $2/3$, and 1, and both f and f' .

17.5.4 Derive

$$\int_0^a \frac{f(x)}{\sqrt{2ax - a^2}} dx$$

using x equal to 0, and a , and both f and f' . Find the error term.

17.5.5 Derive and check $G(s)$

$$\int_0^{3h} f(x) dx = \frac{3h}{80} [13f(0) + 27f(h) + 27f(2h) + 13f(3h)] \\ + \frac{3h^2}{40} [f'(0) - f'(3h)] + \frac{9h^7}{11,200} y^{(7)}(\theta)$$

17.5.6 Derive and check $G(s)$

$$\int_0^{4h} f(x) dx = \frac{2h}{945} \{305[f(0) + f(4)] + 604[f(1) + f(3)]\} \\ - 2h\{27[f'(4) - f'(0)] + 96[f'(3) - f'(1)]\} + \frac{52h^9}{99,225} y^{(9)}(\theta)$$

17.6 BIRKHOFF INTERPOLATION AND FORMULAS

A natural generalization of the previous methods of interpolation is to allow the function and a variable number of derivatives to be used as data at each point; we can use $f(x_i), f'(x_i), \dots, f^{(k_i-1)}(x_i)$ which are k_i consecutive derivatives (0, 1, ..., $k_i - 1$) at the point x_i . This we shall call the Birkhoff¹ interpolation.

The methods used before show that we can find the formula by the method of undetermined coefficients. The error term will turn out to be

$$[(x - x_1)^{k_1}(x - x_2)^{k_2} \cdots (x - x_{n+1})^{k_{n+1}}] \frac{f^{(m)}(\theta)}{m!}$$

where $m = k_1 + k_2 + \cdots + k_{n+1}$.

When we try to find formulas using these same data we are led, in the direct method, to a system of equations whose matrix is related to the Vandermonde, and this can again be inverted by the same technique of finding the sampling polynomials. The existence of the inverse shows that the determinant of the system of equations is not zero; therefore we can indeed find the interpolating polynomials and the formulas of the desired form (provided we can find the moments m_k of the operator).

As an example of the method and the kinds of formulas that can exist, consider

$$\int_{-1}^1 f(x) dx = w_{-1}f(-1) + w_0f(0) + w_1f(1) + w'_0f'(0) + w'_1f'(1) + w''_1f''(1)$$

The coefficients of the defining equations can be arranged as in Table 17.6.

¹ George David Birkhoff (1884–1944).

Producing all the sampling polynomials is awkward, but we can use those that are easy to find (a typical compromise we shall often make).

$$\begin{aligned}\text{I:} \quad & x^2(x-1)^3 = x^5 - 3x^4 + 3x^3 - x^2 + 0x + 0 \\ \text{II:} \quad & (x+1)x(x-1)^3 = x^5 - 2x^4 + 0x^3 + 2x^2 - x + 0 \\ \text{III:} \quad & (x+1)(x^2)(x-1)^2 = x^5 - x^4 - x^3 + x^2 + 0x + 0\end{aligned}$$

The coefficients are in the *respective* columns headed I, II, and III. Multiplying the equations by these numbers and adding, we get the three equations

$$\begin{aligned}\text{I:} \quad & -2/3 - 6/5 = -8w_{-1} \quad \text{thus } w_{-1} = 7/30 \\ \text{II:} \quad & 4/3 - 4/5 = -w'_0 \quad \text{thus } w'_0 = -16/30 \\ \text{III:} \quad & 2/3 - 2/5 = 4w''_1 \quad \text{thus } w''_1 = 2/30\end{aligned}$$

Table 17.6

$f(x)$	m_k	w_{-1}	w_0	w_1	w'_0	w'_1	w''_1	I	II	III
1	2	1	1	1	0	0	0	0	0	0
x	0	-1	0	1	1	1	0	0	-1	0
x^2	2/3	1	0	1	0	2	2	-1	2	1
x^3	0	-1	0	1	0	3	6	3	0	-1
x^4	2/5	1	0	1	0	4	12	-3	-2	-1
x^5	0	-1	0	1	0	5	20	1	1	1

From these three, the other constants are easily found, and we have the formula

$$\int_{-1}^1 f(x) dx = \frac{1}{30} [7f(-1) + 16f(0) + 37f(1) - 16f'(0) - 14f'(1) + 2f''(1)]$$

It should now be apparent that we can find formulas of a wide class—all we need to know are the moments m_k of the operator we are using, and the rest is tedious algebra. The sample points need not be in the range of integration.

PROBLEMS

17.6.1 Derive

$$\int_0^1 f(x) dx = \frac{f(1) + f(0)}{2} - \frac{f'(1) - f'(0)}{10} + \frac{f''(1) + f''(0)}{120} \quad E_7 = -1/20$$

and check $G(s)$.

17.6.2 Derive

$$\int_0^1 f(x) dx = \frac{f(1)+f(0)}{2} - \frac{3}{28} [f'(1)-f'(0)] + \frac{1}{84} [f''(1)+f''(0)] \\ + \frac{1}{1,680} [f'''(1)-f'''(0)] \quad E_9 = 1/70$$

and check $G(s)$.

17.6.3 Find

$$\int_0^\infty e^{-x} f(x) dx = a_0 f(0) + a_1 f(4) + b_0 f'(0) + b_1 f'(4) + c_0 f''(0) + c_1 f''(4)$$

17.6.4 Derive

$$-\int_0^1 (\ln x) f(x) dx = \frac{15}{16} f(0) + \frac{1}{16} f(1) + \frac{3}{16} f'(0) + \frac{7}{288} f''(0)$$

17.6.5 Derive

$$\int_0^\infty e^{-x} f(x) dx = \frac{1}{4} f(0) + \frac{3}{4} f(2) - \frac{1}{2} f'(0) - \frac{1}{2} f''(0)$$

17.7 AN EXAMPLE OF A
NONINTERPOLATORY FORMULA

From the Birkhoff interpolation one is apt to draw the conclusion that almost any reasonable set of information can be used as a basis for interpolation. This is false. The simple example of three equally spaced points with both the function and second derivatives (position and acceleration, if you wish) does *not* determine a quintic (fifth-degree) polynomial. To prove this, let the points be $-1, 0$, and 1 . The function and defining equations are:

$$\begin{aligned} f(x) &= A_0 + A_1 x + A_2 x^2 + A_3 x^3 + A_4 x^4 + A_5 x^5 \\ f(-1) &= A_0 - A_1 + A_2 - A_3 + A_4 - A_5 \\ f(0) &= A_0 \\ f(1) &= A_0 + A_1 + A_2 + A_3 + A_4 + A_5 \\ f''(-1) &= 2A_2 - 6A_3 + 12A_4 - 20A_5 \\ f''(0) &= 2A_2 \\ f''(1) &= 2A_2 - 6A_3 + 12A_4 - 20A_5 \end{aligned}$$